YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT

A Constituent Unit of Yenepoya (Deemed to be University)

A PROJECT REPORT ON

A.C.E

ASPIRE.INSPIRE.CONNECT

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V SEM BACHELOR OF COMPUTER APPLICATION

BDA,DS,BI

AI,ML,RO

DEVOPS

UNDER THE GUIDANCE OF

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DEPARTMENT OF COMPUTER SCIENCE

2025

Abstract

In today's rapidly evolving global economy, students face increasing challenges in choosing suitable career paths due to the fast pace of technological advancement and the emergence of new industries. Traditional career counseling methods, which rely on manual assessments and limited human interaction, are often inadequate to address the complex and dynamic career landscape of the 21st century. To overcome these limitations, the project **A.C.E. (Aspire, Connect, Evolve)** introduces an **AI-powered web application** designed to provide comprehensive, personalized, and data-driven career guidance for students.

The main objective of A.C.E. is to bridge the gap between education and employability by integrating artificial intelligence, data analytics, and real-time market insights into a single, accessible digital platform. The system evaluates each user's interests, academic performance, and personality traits to recommend suitable career options and skill-development roadmaps. In addition, it includes verified databases of institutions, scholarships, and government programs to ensure authenticity and protect students from fraudulent sources.

The platform also facilitates collaboration among students, teachers, and parents through a shared dashboard that promotes informed and transparent decision-making. Built using modern web technologies such as **HTML, CSS, JavaScript, React.js, Node.js, and MongoDB**, A.C.E. delivers a responsive, interactive, and scalable user experience. The integration of AI modules using **Python and machine learning libraries** enhances the accuracy and adaptability of career recommendations.

This project contributes to the modernization of the education system by transforming career guidance into a continuous, data-driven, and inclusive process. Through A.C.E., students gain access to free and reliable digital mentorship, empowering them to make informed choices that align with both their aspirations and global workforce trends. Ultimately, A.C.E. stands as a step toward building a smarter, equitable, and future-ready generation of learners.

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Books and Journals

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Appendices (if any)

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Additional Charts or Data

Chart 1: Global Youth Unemployment Rate (2020–2025)

Year	Global Youth Unemployment (%)	Source
2020	14.9	International Labour Organization (ILO)
2021	14.4	ILO World Employment Report
2022	13.9	World Bank Global Employment Data
2023	13.6	OECD Labour Outlook
2024	13.2	World Economic Forum Future of Jobs Report
2025	*Projected: 12.8*	A.C.E. Research Estimate

Interpretation:
The data demonstrates a slow but steady global decline in youth unemployment, largely influenced by digital transformation, online education, and the integration of AI-enabled guidance tools. A.C.E. aims to accelerate this positive trend by bridging the education-to-employment gap.

Chart 2: Skill Demand by Sector (2025 Projection)

Sector	Skill Demand (%)	Emerging Skills
Artificial Intelligence & Data Science	22%	Python, Machine Learning, Data Visualization
Healthcare & Biotechnology	18%	Bioinformatics, Health Data Analytics
Renewable Energy	14%	Green Tech Design, Environmental Modelling
Business & Management	12%	Data-Driven Decision Making, Leadership
Creative & Media Technology	10%	UX/UI, Digital Marketing, AR/VR
Others	24%	Multidisciplinary and Hybrid Roles

Interpretation:
A.C.E. integrates this data into its AI algorithm to ensure real-time alignment between student interests and emerging skill demands across sectors.

Chart 3: User Growth and Adoption Targets for A.C.E. (Year 1)

Quarter	Planned Institutions Onboarded	Estimated Student Users	Data Insights Generated
Q1	20	10,000	5,000 profiles analyzed
Q2	50	25,000	15,000 insights generated
Q3	100	50,000	30,000 insights generated
Q4	200	1,00,000	70,000 insights generated

Interpretation:
These projections illustrate the scalability and community impact potential of A.C.E. as a public digital infrastructure project.

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Introduction

- Background of your topic

2.1 Background of the Topic

In the 21st century, the landscape of education and employment has undergone a fundamental transformation. Globalization, automation, and technological advancements have redefined how people learn, work, and grow in their careers. With industries evolving at an unprecedented pace, new professions are emerging faster than ever before, while traditional roles are being reshaped or replaced by technology. This dynamic environment has created a growing need for **structured, data-driven career guidance systems** that can help students navigate the complexities of modern career choices.

Unfortunately, despite technological progress, a significant portion of students worldwide still lack access to professional career counseling or reliable guidance resources. In many countries, especially in developing regions, the existing systems for career counseling are either outdated, unavailable, or limited to basic aptitude assessments that fail to capture the multidimensional nature of individual talent. As a result, students often make academic and professional decisions based on incomplete information, peer pressure, or social expectations, rather than personal interests and evolving market demands.

Studies conducted by the **World Economic Forum (2023)** and **UNESCO (2022)** highlight the widening skills gap between what is taught in educational institutions and what is required by employers. The mismatch between learning outcomes and employability has led to an alarming rise in **youth unemployment and underemployment**, especially among recent graduates. Many young individuals enter the workforce without a clear understanding of where their skills fit, resulting in wasted potential and economic inefficiency.

Moreover, with the rise of digital learning platforms and online education, students are exposed to vast information but lack the ability to filter or contextualize it in terms of their career goals. The absence of centralized and trustworthy digital guidance tools leaves room for misinformation, unverified institutions, and fraudulent scholarship schemes. Consequently, many students fall prey to misleading advertisements or unaccredited programs.

To address these persistent challenges, there is an urgent need for an **integrated, AI-powered platform** that not only provides personalized career recommendations but also educates and empowers students to make informed decisions. This is where **A.C.E. (Aspire, Connect, Evolve)** finds its relevance. Designed as an intelligent web application, A.C.E. bridges the gap between education and employment by combining **artificial intelligence, analytics, and real-time career data** to deliver holistic, personalized, and secure career guidance to students, educators, and policymakers alike.

A.C.E. is more than a web application—it represents a **public digital infrastructure for career development**. It democratizes access to information and ensures that every student, irrespective of background, has the opportunity to understand, plan, and pursue a meaningful

career path. The system's goal is to bring career guidance into the mainstream of education, making it an integral part of academic growth and lifelong learning.

Purpose of the Web Application

The primary objective of the **A.C.E. web application** is to revolutionize how students make academic and professional decisions through a structured, evidence-based, and user-friendly digital platform. The purpose extends beyond individual counseling; it is to **create an ecosystem** that connects students, teachers, parents, and policymakers in one collaborative network focused on career development and employability.

The A.C.E. platform serves as a **personalized career guidance system**, utilizing artificial intelligence and data analytics to evaluate users' interests, skills, aptitudes, and personality traits. Based on this comprehensive profile, the platform recommends suitable career paths, courses, and skill development programs. The AI engine is trained on global job market data, ensuring that career suggestions are not only relevant to the user's capabilities but also aligned with future trends and demand in the labor market.

In addition, the platform integrates several functional modules to enhance usability and impact:

Career Recommendation Engine: Uses AI to analyze user profiles and suggest optimal career choices with actionable steps.

Educational Roadmap: Provides detailed pathways, including subjects, certifications, and extracurricular skills required for the selected career.

Real-Time Market Analytics: Keeps students updated with industry trends, emerging job roles, and required competencies.

Institution Database: Lists verified colleges, universities, and training institutions, ensuring students only access legitimate and accredited sources.

Anti-Fraud Detection: The system flags potential scams and misleading information through AI-based verification algorithms.

Parental and Teacher Dashboard: Facilitates transparency and collaboration, enabling stakeholders to guide students with evidence-based insights.

The platform's design focuses heavily on **accessibility and inclusivity**. A.C.E. is a free-to-use application intended primarily for students in government or public education systems who may not otherwise have access to quality guidance. The system's user interface is simple, mobile-friendly, and adaptable for multilingual access to ensure maximum reach.

Beyond direct user benefits, the A.C.E. system supports **data-driven policymaking**. It generates anonymous analytical reports highlighting trends in student preferences, skill gaps, and emerging interest areas. Policymakers and education administrators can utilize these insights to reform curricula, allocate resources, and design programs that prepare youth for the future of work.

Ultimately, the purpose of A.C.E. is to create a **bridge between education and employment**—to guide students from the classroom to the workplace through continuous mentorship, credible information, and technology-enabled support.

Why this project is useful

The **A.C.E. project** provides multiple layers of value to individuals, educational institutions, and society at large. Its usefulness can be analyzed through three perspectives — **educational, technological, and social impact.**

A. Educational Usefulness

Education today is increasingly focused on skill-based learning and employability. A.C.E. supports this transformation by helping students understand the relevance of their academic choices. Instead of choosing subjects blindly, learners can explore how each course aligns with specific career paths and real-world applications.

The platform also assists educators by offering analytical insights into students' interests and readiness levels. Teachers can use this information to tailor guidance sessions, mentor students, and identify potential career mismatches early. This not only enhances academic engagement but also promotes lifelong learning habits among students.

B. Technological Usefulness

Technologically, A.C.E. showcases the potential of **AI-driven education systems**. The application integrates natural language processing, machine learning, and predictive analytics to provide intelligent, adaptive recommendations. Unlike static systems, A.C.E. learns continuously from user feedback, ensuring that the quality of guidance improves over time.

The platform's secure cloud-based architecture enables scalability, allowing it to serve millions of students simultaneously without compromising performance. Its modular design ensures flexibility — meaning that it can be easily integrated into existing school management systems or government education portals.

Furthermore, A.C.E. demonstrates the **ethical application of artificial intelligence** in education. The system respects data privacy, transparency, and fairness, aligning with UNESCO's AI Ethics Framework. Its emphasis on verified information and anti-fraud features makes it a safe digital environment for students to explore their futures.

C. Social Usefulness

From a social standpoint, A.C.E. contributes to building **equity and inclusion** in education. Many students from rural or economically disadvantaged backgrounds lack access to professional counselors or career workshops. A.C.E. breaks this barrier by providing free, accessible, and trustworthy career guidance to anyone with an internet connection.

By bridging this digital divide, A.C.E. not only empowers individuals but also supports national development by reducing unemployment and ensuring that the workforce is aligned with future market needs. When implemented at scale, the platform can help governments identify skill shortages, plan workforce development initiatives, and create more targeted educational reforms.

The platform also promotes **ethical and informed career decision-making.** By providing transparent, verified, and research-based information, A.C.E. helps students avoid educational scams and exploitative schemes. This ensures that career planning becomes an empowering process rather than a risky or confusing one.

D. Broader Impact

The usefulness of A.C.E. extends into **policy-level innovation**. Governments and NGOs can use aggregated data from the system to understand regional employment trends, assess the effectiveness of existing education programs, and plan interventions for sectors with low participation or skill deficits.

On a global level, A.C.E. aligns with the **United Nations Sustainable Development Goals (SDGs)**, particularly:

> **SDG 4:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
>
> **SDG 8:** Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.
>
> **SDG 10:** Reduce inequality within and among countries.

By combining education, technology, and social development, A.C.E. contributes directly to building a more equitable and future-ready society.

Conclusion of Introduction

The A.C.E. (Aspire, Connect, Evolve) web application stands as a forward-thinking initiative that addresses one of the most critical needs in today's education system — the lack of accessible, intelligent, and trustworthy career guidance. It merges the precision of artificial intelligence with the compassion of human-centered design to empower students in making well-informed career decisions.

By democratizing access to career information, connecting all educational stakeholders, and using technology ethically, A.C.E. transforms the vision of **"Where Ambition Meets Direction"** into a tangible reality. This project is not merely an innovation in web technology — it is a movement toward redefining how future generations learn, grow, and succeed.

System Requirements

a) **Hardware Requirements**
b) Software Requirements
(Example: HTML, CSS, JavaScript, Bootstrap, Database, VS Code, etc.)

Hardware Requirements

System Used : MacBook M2 2024 , Asus RiGor

Software Requirements

The **A.C.E. (Aspire, Connect, Evolve)** platform is a web-based application that integrates Artificial Intelligence (AI), Data Analytics, and an interactive user interface for real-time career guidance. The system is developed using modern, scalable, and open-source technologies to ensure performance, accessibility, and maintainability. The software requirements include all the tools, frameworks, and environments essential for the design, development, testing, and deployment of the system.

1. Front-End Technologies

Technology	Purpose / Description
HTML5 (Hypertext Markup Language)	Used to structure and design the basic layout of the web pages and define the content hierarchy.
CSS3 (Cascading Style Sheets)	Provides styling, layout, and visual consistency across the platform, ensuring a clean and responsive design.
Bootstrap 5	A responsive front-end framework used to create mobile-friendly interfaces with predefined UI components and grids.
JavaScript (ES6)	Enables interactivity, event handling, and dynamic content updates in the user interface.
React.js *(or Angular/Next.js)*	Used as the main front-end framework for building a modular, reusable, and fast-performing user interface.

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2. Back-End Technologies

Technology	Purpose / Description
Node.js	Provides the runtime environment for executing JavaScript code on the server side, allowing scalable API development.
Express.js	A Node.js framework used to build RESTful APIs for communication between the front-end and the database.
Python (AI	Handles the AI and data analytics functions including career

Technology	Purpose / Description
Module)	recommendation algorithms and data preprocessing.
Flask / FastAPI (optional)	Used for integrating Python-based AI models with the main server for efficient model inference.

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3. Database Management System (DBMS)

Technology	Purpose / Description
MongoDB / MySQL	Acts as the main database for storing user data, profiles, and AI model results. MongoDB supports flexibility with NoSQL document structures, while MySQL can be used for relational table-based data storage.
Mongoose (for MongoDB)	Simplifies database schema design and validation within Node.js applications.
SQLAlchemy (if MySQL used)	ORM library for Python, used to query and manage relational data efficiently.

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4. Artificial Intelligence and Data Analytics Tools

Tool / Library	Purpose / Description
Scikit-learn	For implementing machine learning algorithms such as classification and clustering to generate career recommendations.
Pandas & NumPy	For data handling, statistical analysis, and transformation of large datasets for training the recommendation model.
TensorFlow / PyTorch	Used for building and training deep learning models that enhance prediction accuracy.
Matplotlib / Seaborn	For visualization of data trends, model performance, and dashboard analytics.

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5. Development Tools and IDEs

Tool	Purpose / Description
Visual Studio Code (VS Code)	Main integrated development environment used for writing, debugging, and testing code across front-end and back-end.
Git & GitHub	Version control system and repository used to track project development, manage branches, and enable team collaboration.
Postman	For testing REST APIs and verifying client-server communication.

Tool	Purpose / Description
Jupyter Notebook	Used for AI model experimentation, training, and documentation.
Docker (optional)	For containerizing the web application to simplify deployment and ensure consistency across environments.

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6. Deployment and Hosting Platforms

Platform / Tool	Purpose / Description
AWS / Microsoft Azure / Google Cloud Platform (GCP)	Cloud services used to host the web application, ensuring scalability and reliability.
Firebase (optional)	Used for hosting front-end assets and managing real-time data if needed.
Nginx / Apache	Web servers for managing requests and serving content securely to users.
Domain and SSL Certificate	To register and secure the website, ensuring encrypted communication between users and the server.

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7. Testing and Quality Assurance Tools

Tool / Framework	Purpose / Description
Jest / Mocha	JavaScript testing frameworks for validating front-end and back-end logic.
Selenium / Cypress	Used for end-to-end testing of user interface interactions and workflow validation.
PyTest (for AI models)	For testing machine learning functions and ensuring correct prediction outputs.
Lighthouse (Google)	For testing web performance, accessibility, and SEO optimization.

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8. Operating System and Browser Compatibility

Component	Description
Operating System (Development)	Windows 10/11, macOS, or Linux (Ubuntu)
Browser Compatibility	Chrome, Microsoft Edge, Firefox, Safari
Server Environment	Node.js Runtime Environment and Python Environment

Component	Description
	(Anaconda / Virtual Environment)

Existing System

- What currently exists?
- Limitations/problems in existing system

4. Existing System

4.1 What Currently Exists

Career guidance plays a vital role in shaping the educational and professional pathways of students. However, the majority of existing career guidance systems — especially in developing nations — are **traditional, fragmented, and underdeveloped**. Most educational institutions still rely on manual or semi-digital systems that depend on career counselors, aptitude tests, or generic online tools.

The **current career guidance ecosystem** can generally be categorized into three types:

Manual Counseling Systems
In many schools and colleges, career counseling is conducted through periodic sessions led by teachers, mentors, or external counselors. These sessions are typically one-time interactions where students are provided with basic information about available career options. The process is subjective and highly dependent on the counselor's experience and knowledge. As a result, there is little standardization or personalization.

Private Career Counseling Agencies
Several private organizations offer paid career guidance services, using psychometric assessments and one-on-one consultations. While these platforms are more structured, their services are often **expensive and inaccessible** to students from middle- or low-income backgrounds. Additionally, most private services focus on short-term counseling and do not provide continuous support or follow-up.

Online Career Guidance Portals
With the advancement of technology, various websites and applications such as *LinkedIn Learning, Coursera Career Academy,* and *India's National Career Service (NCS)* provide career information and job listings. However, these platforms primarily focus on job-seeking and training rather than personalized guidance. They lack **AI-driven recommendations**, **psychometric integration**, and **interactive collaboration** among students, teachers, and parents.

Institutional Career Guidance Programs
A few universities and educational boards have initiated online career assessment portals or "Career Corners" on their websites. However, these are limited in functionality, providing static lists of professions, courses, and colleges without adaptive recommendations or real-time data. They often fail to accommodate the rapidly changing job market and global skill requirements.

Overall, while there are numerous resources available, they operate **independently**, without integration or personalization. Students are often required to browse multiple platforms for different needs—such as aptitude tests, college searches, scholarship databases, and job updates—leading to confusion and poor decision-making.

4.2 Limitations / Problems in Existing System

Despite the availability of several online and offline options, the existing systems for career guidance have **multiple limitations** that hinder their effectiveness and inclusivity. The key challenges are outlined below:

1. Lack of Personalization

Most traditional career counseling systems use a "one-size-fits-all" approach. Students are grouped into generic categories based on academic scores or simple aptitude tests. These methods do not consider **individual interests, personality traits, learning styles, or long-term aspirations.** As a result, many students receive advice that is irrelevant or uninspiring, leading to poor motivation and misaligned career choices.

2. Limited Accessibility

Professional career counseling services are often **available only in urban areas or private schools**, leaving rural and underprivileged students without access. Digital solutions that exist are either subscription-based or require advanced computer literacy, making them inaccessible to many learners.

3. Absence of Real-Time Labor Market Data

Existing platforms rarely integrate **live data from global job markets** or emerging trends in industries like AI, clean energy, or data analytics. As a result, the recommendations provided are based on outdated career structures rather than future-oriented insights. This lack of adaptability leads to a **skills mismatch** between graduates and employers.

4. Fragmented Information Sources

Students currently have to visit multiple websites for career tests, college admissions, scholarships, and internship opportunities. These scattered systems cause confusion and **information overload**, making it difficult for students to identify reliable data or plan coherent career paths.

5. High Cost of Private Platforms

Private career guidance tools and consultancy services can cost anywhere between ₹2,000–₹10,000 per student, depending on the package. This cost creates an **economic barrier** for most students in public education systems, thereby widening the gap between privileged and underprivileged learners.

6. Lack of Continuous Guidance

Existing systems tend to offer **one-time counseling** rather than continuous mentoring or progress tracking. Once a student receives initial advice, there is usually no follow-up mechanism to measure progress or adapt the guidance as the student develops new skills or interests.

7. Absence of Parental and Teacher Involvement

In many existing systems, **career guidance is treated as an individual task**, ignoring the influence of teachers and parents. This isolation leads to miscommunication and a lack of collective support. Students often struggle to justify their career choices to their families or educators due to limited collaboration tools.

8. Outdated Technology and Static Interfaces

Many web-based platforms still operate with outdated technology, using static pages with minimal interactivity. They fail to incorporate AI, data analytics, or personalization features that today's students expect. Moreover, the design and user experience are often not optimized for mobile devices, which limits accessibility.

9. Lack of Verification and Security

The online career guidance ecosystem is also plagued by **unverified institutions, fake scholarship offers, and fraudulent courses**. Students who rely on unregulated websites risk wasting time, money, and effort. Existing systems lack mechanisms to verify data sources or detect scams using AI.

10. No Integration Between Education and Employment Systems

There is a fundamental disconnect between educational institutions and the employment sector. Most guidance platforms do not communicate with job databases or employer networks. As a result, even after following a recommended educational path, students may find that their qualifications do not match industry needs.

4.3 Summary of Limitations

Category	Key Problem	Impact on Students
Personalization	One-size-fits-all counseling methods	Misguided or irrelevant career paths
Accessibility	Services limited to urban/private schools	Exclusion of rural or low-income students
Data Relevance	Lack of real-time job market insights	Skills mismatch and unemployment
Information	Fragmented across multiple sources	Confusion and decision fatigue
Cost	Paid services only	Inequality in access to guidance
Continuity	No progress tracking	Lack of long-term career planning
Collaboration	Minimal parent/teacher involvement	Weak decision-making support
Technology	Outdated platforms	Poor user engagement
Security	No scam detection or data verification	Risk of fraud and misinformation
Integration	No link between education and employment	Ineffective transition to workforce

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4.4 Need for Improvement

Given these shortcomings, there is an urgent need for a **centralized, intelligent, and collaborative digital career guidance platform** that:

Uses **AI and data analytics** to personalize recommendations.

Offers **free and equal access** to all students, regardless of background.

Provides **real-time labor market insights** and verified data.

Encourages **continuous guidance** involving students, teachers, and parents.

Detects **fraudulent educational sources** using smart verification algorithms.

The **A.C.E. (Aspire, Connect, Evolve)** web application has been designed to address each of these gaps systematically — creating a modern, accessible, and ethical solution for 21st-century learners.

5. Proposed System

5.1 Your Solution

The proposed solution, **A.C.E. (Aspire, Connect, Evolve)**, is an **AI-powered web application** developed to provide a **comprehensive, personalized, and data-driven career guidance system** for students, educators, and parents. Unlike traditional or fragmented platforms, A.C.E. integrates multiple functionalities into a single unified ecosystem that focuses on empowerment, collaboration, and real-time intelligence.

The primary goal of A.C.E. is to bridge the critical **career guidance gap** that exists between education and employability. By leveraging artificial intelligence, data analytics, and machine learning algorithms, A.C.E. delivers tailored career recommendations based on each student's interests, abilities, academic background, and personality traits. The platform continuously evolves with market data, ensuring that students are always guided toward careers that are **relevant, achievable, and future-proof**.

A.C.E. operates through three fundamental pillars:

Aspire: Encourages students to explore their interests and set meaningful career goals. Through psychometric assessments and interactive profiling, the system identifies each user's aptitude, strengths, and motivation areas.

Connect: Builds a bridge between students, teachers, and parents by creating a collaborative environment. Teachers can track student progress, parents can stay informed, and administrators can use aggregated data for policy decisions.

Evolve: Promotes lifelong learning and adaptability by recommending skill-building courses, certifications, and real-world experiences that align with the student's chosen path.

The A.C.E. platform is designed as a **cloud-based and modular web system** consisting of the following components:

User Dashboard: Personalized space for each student showing career matches, recommendations, and progress indicators.

AI Recommendation Engine: Core module that analyzes user data and market trends to generate accurate and dynamic suggestions.

Data Analytics Dashboard (for Teachers/Administrators): Provides insights into student interests, regional trends, and skill gaps.

Verified Institution & Scholarship Database: Ensures only authentic and accredited educational opportunities are displayed.

Fraud Detection and Warning System: Uses AI to flag suspicious or unverified institutions and programs.

Parent Access Portal: Allows parents to view guidance reports and assist students in career planning.

The entire system is **interactive, multilingual, and mobile-friendly**, ensuring accessibility for students from diverse educational and geographic backgrounds.

By centralizing all career guidance services on one intelligent platform, A.C.E. ensures that every student can **discover, plan, and pursue** their dream career through reliable and verified information.

5.2 Advantages over Existing System

The **A.C.E. system** provides significant improvements over existing manual and online guidance models by incorporating advanced technologies and a holistic design approach. The key advantages are as follows:

1. Personalization through AI

Unlike traditional systems that provide generic recommendations, A.C.E. uses **artificial intelligence and psychometric profiling** to create individualized career roadmaps. The AI

engine continuously refines its recommendations based on user feedback, academic progress, and job market trends.

2. Unified and Centralized Platform

Instead of navigating multiple fragmented websites for colleges, courses, scholarships, and tests, A.C.E. consolidates all essential resources into one portal. This integrated design improves usability, reduces confusion, and provides an **end-to-end guidance experience**.

3. Accessibility and Inclusivity

A.C.E. is a **free and publicly available platform** designed for accessibility across devices, including mobile phones. It ensures equal access for students from all backgrounds, including those in rural or underprivileged regions who lack access to professional counselors.

4. Real-Time Market Intelligence

The system is connected to **real-time labor market analytics**, enabling students to view trending industries, in-demand skills, and job availability. This ensures that recommendations are not only personalized but also relevant to the **current and future job ecosystem**.

5. Continuous Mentorship and Tracking

A.C.E. offers **long-term guidance** rather than one-time counseling. Progress indicators, performance dashboards, and periodic updates allow students to monitor their development and adjust their plans accordingly.

6. Verified and Secure Information

One of the biggest risks in existing systems is misinformation and fraudulent institutions. A.C.E. includes a **verified institution database** that lists only government-approved and accredited colleges, courses, and scholarship programs. The platform also uses AI algorithms to detect and flag scams, thereby protecting students from exploitation.

7. Data-Driven Decision Support for Educators

Teachers, administrators, and policymakers can use the **analytics dashboard** to identify emerging student interests, curriculum needs, and workforce trends. This data-driven insight enables more informed decision-making and supports education reforms at institutional and policy levels.

8. Parental Collaboration

Unlike most platforms that isolate students in the decision-making process, A.C.E. encourages **family participation**. Parents receive reports and can communicate with counselors through the system, ensuring that career decisions are supported by both data and understanding.

9. Scalable and Sustainable Design

Built on cloud infrastructure with open-source technologies, A.C.E. can scale easily to serve millions of users simultaneously. Its modular architecture allows it to be adapted for different regions, languages, and educational systems without heavy re-engineering.

10. Ethical and Transparent System

A.C.E. adheres to the principles of **ethical AI**, ensuring privacy, fairness, and transparency in all operations. Student data is securely stored and processed only with consent, complying with international data protection standards.

5.3 New Features You Are Adding

A.C.E. introduces several **innovative features** that go beyond what traditional and existing career guidance systems offer. These features represent the next generation of **intelligent educational technology**.

1. AI-Powered Career Recommendation System

The central feature of A.C.E. is an AI engine that analyzes each student's personality, academic performance, and interests to generate a ranked list of potential career options. It uses **machine learning algorithms** trained on real-world data to ensure accurate predictions.

2. Career Readiness Score

Each user receives a dynamic **Career Readiness Score**, which indicates how prepared they are for their selected career path. The score changes based on academic achievements, skill acquisition, and participation in relevant activities, motivating students to stay on track.

3. Real-Time Market Alignment

The platform automatically updates its database with current job trends, salary data, and skill requirements from trusted sources such as the World Economic Forum, LinkedIn, and government labor departments. This ensures students receive **market-aligned recommendations** at all times.

4. Fraud Detection and Warning System

Using natural language processing (NLP) and anomaly detection, A.C.E. identifies **unverified or suspicious educational programs**, warning users before they apply. This feature helps students avoid scams and prioritize verified institutions.

5. Verified Database of Institutions and Scholarships

The web app includes a **government-linked and verified database** of colleges, courses, and scholarships. Students can explore legitimate opportunities with confidence, eliminating the risk of misinformation.

6. Multi-User Collaboration Dashboard

The system allows **multi-role access** for students, teachers, and parents. Teachers can track progress and provide mentorship, while parents can view reports and suggestions, promoting a holistic and transparent guidance process.

7. Personalized Roadmaps and Skill Builder

For every career choice, A.C.E. generates a **customized roadmap** showing subjects, certifications, internships, and skills required. It also integrates with e-learning platforms to suggest relevant online courses.

8. Analytics and Reporting Tools

A.C.E. provides **institution-level analytics** for schools and governments to understand student interests and national skill gaps. This data can be used to improve curriculum design and resource allocation.

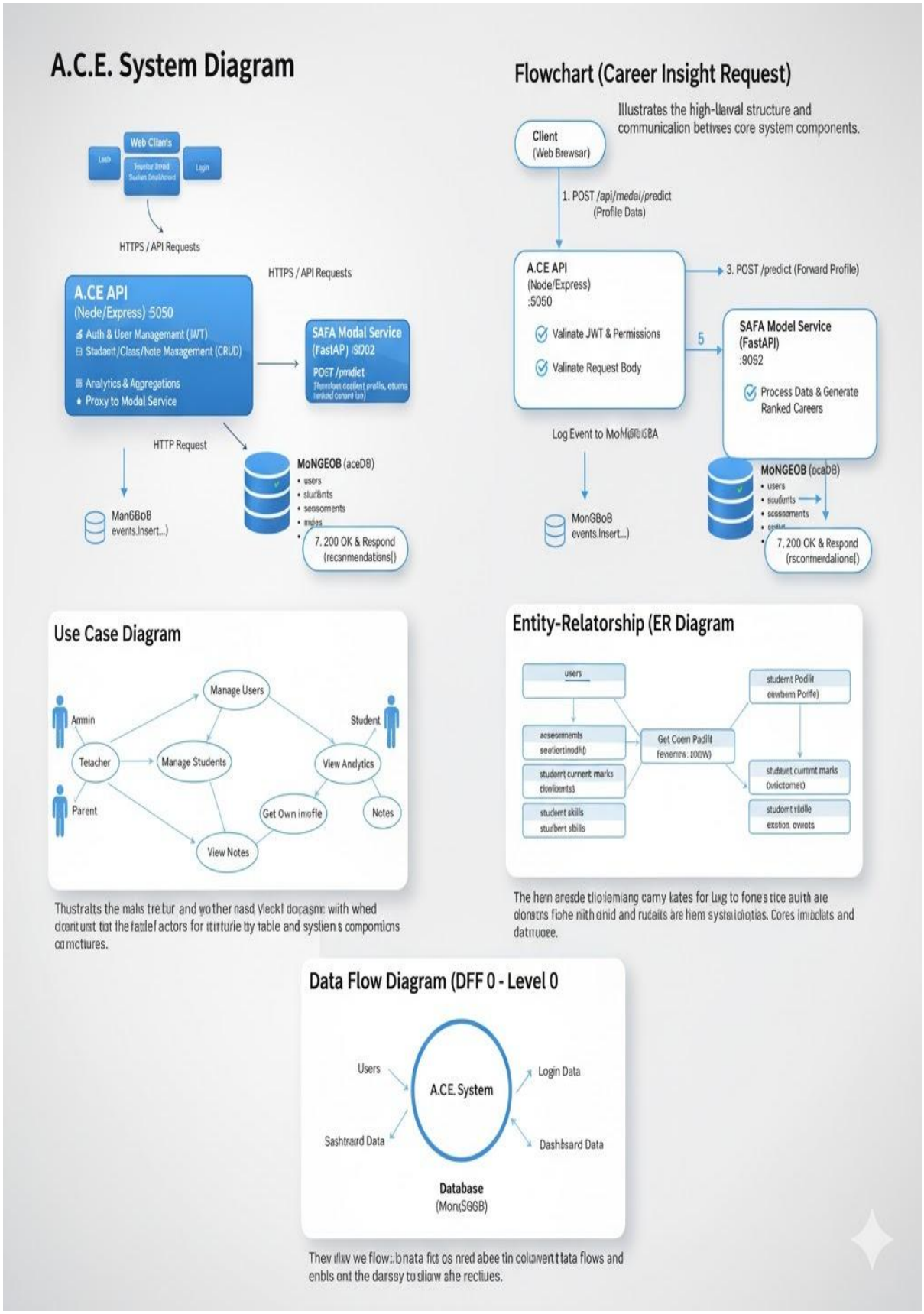
9. Interactive and Responsive Design

The platform is built with modern frameworks such as **React.js and Bootstrap 5**, ensuring that it is fast, user-friendly, and accessible across all devices, including smartphones, tablets, and laptops.

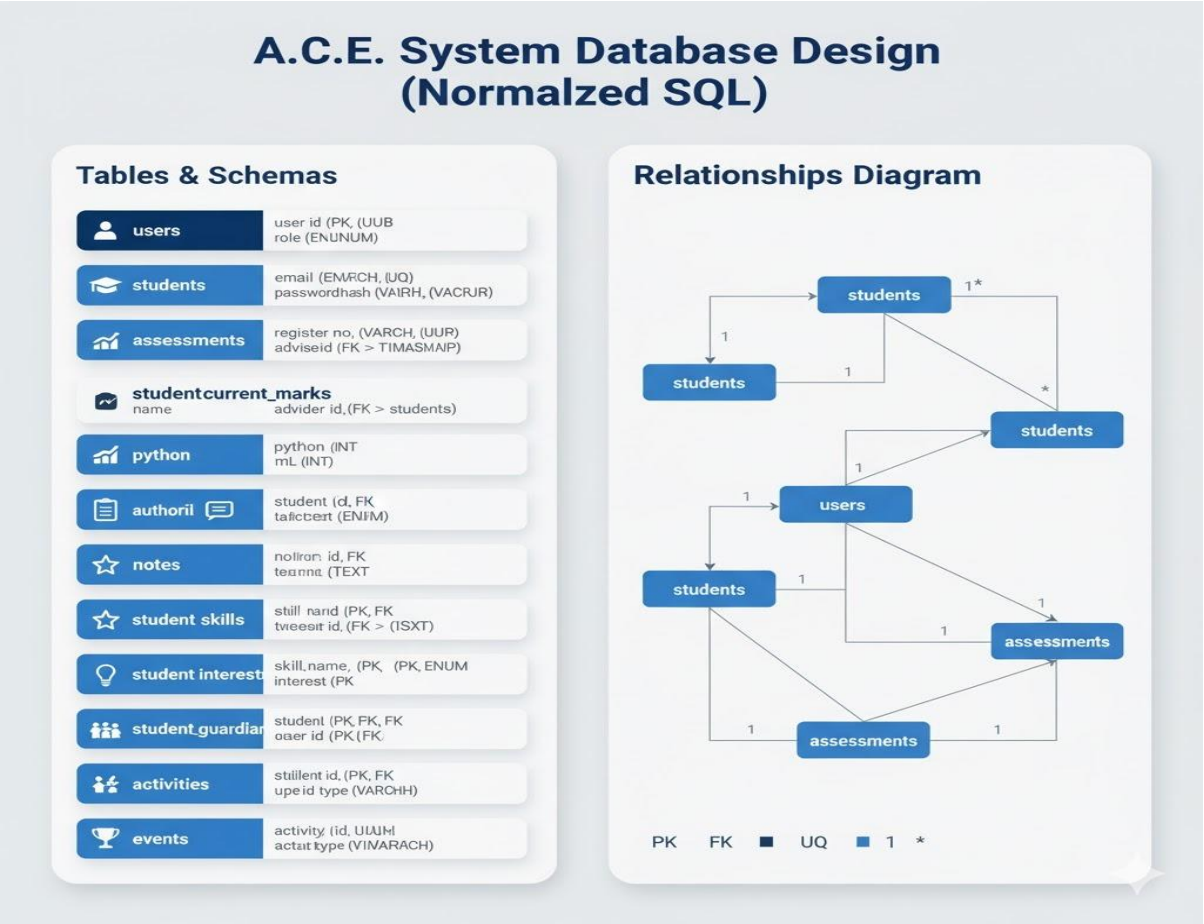
10. Cloud-Based and Scalable Infrastructure

Hosted on **AWS or Azure Cloud**, A.C.E. guarantees 24/7 availability, security, and scalability. This ensures smooth functioning even with large-scale adoption across schools and institutions.

System Design



Database Design



Implementation

Tools and Technologies Used

Category	Technology	Purpose
Frontend	React.js / Next.js	Build interactive student, parent & teacher dashboards
Backend	Node.js (Express)	Core API, authentication, user management
Database	MongoDB / Firebase	Store profiles, progress data, institution database
AI/ML	Python (Scikit-Learn, NLP)	Career matching, scam detection, interest-skill analysis
Cloud Platform	AWS / Google Cloud	Hosting, serverless functions, storage
Authentication	Firebase Auth / JWT	Secure role-based login
Analytics	Power BI / Tableau	Career trends & stakeholder analytics (policy insights)
Other	Git & GitHub	Version control and collaboration

Main Modules Explanation

1. User Management Module

Handles registration & role-based login (Student, Parent, Teacher)

Tracks student profile progress & achievements

Key Features

Secure authentication

Profile dashboard

Account recovery

---

2. Career Recommendation Engine

Uses student interests, personality & market trends to recommend suitable careers.

Functions

Interest assessment

Skill gap analysis

AI-powered career match ranking

---

3. Roadmap & Progress Tracking Module

Displays personalized skill & subject roadmap

Tracks career readiness score in real-time

Rewards system for milestones

---

4. Verified Institution & Scholarship Database

Shows verified colleges & scholarships

AI flags suspicious institutions

Government/NGO integration (future)

---

5. Admin & Analytics Dashboard

Used by teachers, parents & policymakers to monitor learning & trends.

Insights include

Most popular careers

Skills gap by student group

Real-time performance analytics

Code

```
// ==================================================
// ✅ A.C.E Backend — Database Connection
// ==================================================
const mongoose = require("mongoose");

const connectDB = async() => {
```

```
    try {
        const conn = await mongoose.connect(process.env.MONGO_URI);
        console.log(`✅ MongoDB Connected: ${conn.connection.host}`);
    } catch (err) {
        console.error(`❌ Error connecting to MongoDB: ${err.message}`);
        process.exit(1);
    }
};

module.exports = connectDB;

const Student = require('../models/Student');

// Add or update student activity
const addActivity = async(req, res) => {
    try {
        const { name, standard, year, type, description } = req.body;

        let student = await Student.findOne({ name });
        if (!student) {
            student = new Student({ name, standard, activities: [] });
        }

        student.activities.push({ year, type, description });
        await student.save();
```

```
            res.status(201).json({ message: '✅ Activity added successfully', student });
        } catch (error) {
            res.status(500).json({ error: error.message });
        }
    };

    // Get student by name
    const getStudent = async(req, res) => {
        try {
            const student = await Student.findOne({ name: req.params.name });
            if (!student) return res.status(404).json({ message: 'Student not found' });

            res.json(student);
        } catch (error) {
            res.status(500).json({ error: error.message });
        }
    };
```

```
import yaml
import numpy as np

class CareerEngine:
    def __init__(self, config_path="config/careers.yaml"):
        with open(config_path, "r") as f:
```

```
        self.config = yaml.safe_load(f)


    def score(self, profile):
        results = []
        for career in self.config["careers"]:
            academic_score = self._weighted_score(
                career.get("subject_weights", {}),
                profile["academics"].get("marks", {})
            )
            skill_score = self._weighted_score(
                career.get("pro_skill_weights", {}),
                profile["skills"].get("professional", {})
            )
            final_score = (
                0.4 * academic_score +
                0.4 * skill_score +
                0.2 * np.random.uniform(0.7, 1.0)  # small variation
            )
            results.append({
                "careerName": career["name"],
                "confidence": round(final_score, 2)
            })
        results.sort(key=lambda x: x["confidence"], reverse=True)
        return results[:3]
```

```
    def _weighted_score(self, weights, values):
        if not weights: return 0
        total, score = 0, 0
        for k, w in weights.items():
            v = values.get(k, 0)
            total += w
            score += w * (v/100 if v > 1 else v)
        return score / total if total else 0

from fastapi import FastAPI
from pydantic import BaseModel
from datetime import datetime
from engine import CareerEngine

app = FastAPI(title="SAFA Career Model", version="1.0")

# Initialize the engine
engine = CareerEngine(config_path="config/careers.yaml")


# ----- Models -----
class StudentProfile(BaseModel):
    studentId: str
    academics: dict
    skills: dict
```

```
    interests: list
    extracurriculars: list = []
    teacherFeedback: dict = {}


# ----- Routes -----
@app.get("/health")
def health():
    return {"ok": True}


@app.post("/predict")
def predict(profile: StudentProfile):
    result = engine.score(profile.dict())
    return {
        "studentId": profile.studentId,
        "generatedAt": datetime.utcnow().isoformat(),
        "recommendations": result
    }


if __name__ == "__main__":
    import uvicorn
    uvicorn.run(app, host="0.0.0.0", port=8082)
```

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>A.C.E Login | Tap for Tips</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: "Tahoma", sans-serif;
        }

        body {
            height: 100vh;
            overflow: hidden;
            display: flex;
            justify-content: center;
            align-items: center;
            background: #000;
            color: #d4af37;
            position: relative;
        }
        /* === Gold Network Background === */

        canvas#network {
            position: fixed;
            inset: 0;
            z-index: -1;
            background: radial-gradient(circle at top left, #0a0a0a 0%, #000 90%);
        }
        /* === Login Box === */

        .container {
            width: 400px;
            padding: 35px 30px 25px 30px;
            background: rgba(0, 0, 0, 0.85);
```

```
    border: 1px solid #d4af37;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 25px rgba(212, 175, 55, 0.15);
    z-index: 1;
}

.logo img {
    width: 140px;
    border-radius: 10px;
    margin-bottom: 15px;
}

.title {
    font-size: 1.6em;
    font-weight: 600;
    margin-bottom: 4px;
    color: #d4af37;
}

.byline {
    font-size: 0.9em;
    color: #bfa94f;
    margin-bottom: 12px;
    letter-spacing: 1px;
}

.role-buttons {
    display: flex;
    justify-content: space-between;
    margin-bottom: 15px;
}

.role-btn {
    flex: 1;
    background: none;
    border: 1px solid #d4af37;
    color: #d4af37;
    padding: 8px 0;
```

```
    border-radius: 6px;
    margin: 0 5px;
    cursor: pointer;
    font-size: 0.85em;
    transition: 0.3s;
}

.role-btn:hover,
.role-btn.active {
    background: #d4af37;
    color: #000;
}

#error-message {
    color: #ff4d4d;
    font-size: 0.9em;
    min-height: 20px;
    margin-bottom: 10px;
}

form {
    text-align: left;
    display: flex;
    flex-direction: column;
}

label {
    font-size: 0.9em;
    color: #bfa94f;
    margin: 8px 0 4px 3px;
}

.input-wrapper {
    position: relative;
}

input {
    width: 100%;
    background: #000;
```

```
    color: #fff;
    border: 1px solid #d4af37;
    border-radius: 6px;
    padding: 10px;
    font-size: 1em;
    outline: none;
    transition: 0.3s;
    caret-color: #d4af37;
}

input:focus {
    border-color: #b9972e;
}

.eye-icon {
    position: absolute;
    right: 12px;
    top: 50%;
    transform: translateY(-50%);
    width: 22px;
    height: 22px;
    fill: #d4af37;
    cursor: pointer;
    transition: fill .2s, transform .2s;
}

.eye-icon:hover {
    fill: #ffdb58;
    transform: translateY(-50%) scale(1.1);
}

.login-btn {
    margin-top: 15px;
    background: #d4af37;
    color: #000;
    font-weight: bold;
    padding: 10px;
    border: none;
    border-radius: 6px;
```

```
    font-size: 1em;
    cursor: pointer;
    transition: .3s;
}

.login-btn:hover {
    background: #c2a12e;
}

.links {
    margin-top: 12px;
    text-align: center;
    font-size: 0.85em;
}

.links a {
    color: #b2962d;
    text-decoration: none;
    margin: 0 6px;
    transition: .3s;
}

.links a:hover {
    color: #fff;
}
/* === Tips Section === */

.tips-section {
    margin-top: 18px;
    padding-top: 12px;
    border-top: 1px solid rgba(212, 175, 55, 0.3);
    color: #d4af37;
    text-align: center;
    font-size: 0.9rem;
    letter-spacing: 1px;
    text-transform: uppercase;
    cursor: pointer;
    user-select: none;
}
```

```
        .tips-section span {
            display: block;
            font-size: 0.85rem;
            margin-top: 6px;
            color: #ffdb58;
            opacity: 0;
            transition: opacity 1.2s ease;
            font-weight: 600;
        }

        .tips-section:hover strong {
            color: #ffdb58;
            transition: color 0.3s ease;
        }

        footer {
            position: absolute;
            bottom: 12px;
            width: 100%;
            text-align: center;
            color: #777;
            font-size: 0.75em;
            letter-spacing: 0.5px;
        }
    </style>
</head>

<body>
    <canvas id="network"></canvas>

    <div class="container" id="login-box">
        <div class="logo">
            <img src="https://i.postimg.cc/xjy79m1t/Whats-App-Image-2025-10-12-at-13-32-07-
54be146c.jpg" alt="A.C.E Logo">
        </div>
        <div class="title">A.C.E</div>
        <div class="byline">Aspire • Connect • Evolve</div>
```

```
    <div class="role-buttons">
      <button class="role-btn" onclick="selectRole(this)">Student</button>
      <button class="role-btn" onclick="selectRole(this)">Teacher</button>
      <button class="role-btn" onclick="selectRole(this)">Parent</button>
    </div>

    <div id="error-message"></div>

    <form id="loginForm" onsubmit="return validateForm()">
      <label for="username">Username or Email</label>
      <input id="username" type="text" placeholder="Enter your username" required>
      <label for="password">Password</label>
      <div class="input-wrapper">
        <input id="password" type="password" placeholder="Enter your password"
required>
        <svg id="toggleEye" class="eye-icon" viewBox="0 0 24 24"
onclick="togglePassword()">
     <path d="M12 5C7 5 2.73 8.11 1 12c1.73 3.89 6 7 11 7s9.27-3.11 11-7
     c-1.73-3.89-6-7-11-7zm0 10a5 5 0 1 1 0-10 5 5 0 0 1 0 10z"/>
    </svg>
      </div>
      <button type="submit" class="login-btn">Login</button>
    </form>

    <div class="links">
      <a href="#">Forgot Password?</a> | <a href="#">Sign Up</a>
    </div>

    <div class="tips-section" id="tips-section">
      <strong>TIPS:</strong>
      <span id="tip-text"></span>
    </div>
  </div>

  <footer>© 2025 A.C.E Initiative | Concept Prototype</footer>

  <script>
    /* === Role Logic === */
    let selectedRole = null;
```

```
const errorMsg = document.getElementById('error-message');

function selectRole(btn) {
  document.querySelectorAll('.role-btn').forEach(b => b.classList.remove('active'));
  btn.classList.add('active');
  selectedRole = btn.textContent.trim();
  errorMsg.textContent = "";
}

function showError(msg) {
  errorMsg.textContent = msg;
  errorMsg.style.color = "#ff4d4d";
}

/* === Password Toggle === */
const eye = document.getElementById('toggleEye');
const pwd = document.getElementById('password');
let visible = false;

function togglePassword() {
  visible = !visible;
  pwd.type = visible ? "text" : "password";
  eye.innerHTML = visible ?
    `<path d="M1 12c1.9 3.7 6.1 7 11 7 4.9 0 9.1-3.3 11-7-1.9-3.7-6.1-7-11-7-4.9 0-9.1
3.3-11 7zm17.5-4.2l-1.3 1.3A5 5 0 0 1 12 17a5 5 0 0 1-4.9-6.2l-1.6 1.6-1.4-1.4 14-14 1.4 1.4-
2.4 2.4z"/>` :
    `<path d="M12 5C7 5 2.73 8.11 1 12c1.73 3.89 6 7 11 7s9.27-3.11 11-7
   c-1.73-3.89-6-7-11-7zm0 10a5 5 0 1 1 0-10 5 5 0 0 1 0 10z"/>`;
}

/* === Actual Login Function === */
async function validateForm() {
  const email = document.getElementById('username').value.trim();
  const password = document.getElementById('password').value.trim();

  if (!selectedRole) {
    showError("Please select your role.");
    return false;
  }
```

```
if (!email) {
    showError("Please enter your email.");
    return false;
}
if (!password) {
    showError("Please enter your password.");
    return false;
}

try {
    const res = await fetch("/api/auth/login", {
        method: "POST",
        headers: {
            "Content-Type": "application/json"
        },
        body: JSON.stringify({
            email,
            password
        })
    });

    const data = await res.json();

    if (!res.ok) {
        showError(data.message || "❌ Login failed");
        return false;
    }

    // Save user info locally
    localStorage.setItem("aceUser", JSON.stringify(data.user));
    errorMsg.style.color = "#00cc66";
    errorMsg.textContent = "✅ Login successful! Redirecting...";

    // Role-based redirect
    const role = data.user.role;
    setTimeout(() => {
        if (role === "Student") window.location.href = "student-dashboard.html";
        else if (role === "Teacher") window.location.href = "teacher-dashboard.html";
        else if (role === "Parent") window.location.href = "parent-dashboard.html";
```

```
            else showError("❌ Unknown role. Contact admin.");
        }, 1200);

    } catch (err) {
        showError("   Server not reachable. Please try again.");
        console.error(err);
    }
    return false;
}

/* === Gold Particle Background === */
const canvas = document.getElementById("network");
const ctx = canvas.getContext("2d");
let w, h, particles = [];

function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
}
window.addEventListener("resize", resize);
resize();

class Particle {
    constructor() {
        this.x = Math.random() * w;
        this.y = Math.random() * h;
        this.vx = (Math.random() - 0.5) * 0.3;
        this.vy = (Math.random() - 0.5) * 0.3;
        this.size = 1.5;
    }
    move() {
        this.x += this.vx;
        this.y += this.vy;
        if (this.x < 0 || this.x > w) this.vx *= -1;
        if (this.y < 0 || this.y > h) this.vy *= -1;
    }
    draw() {
        ctx.beginPath();
        ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
```

```
        ctx.fillStyle = "rgba(255,215,120,0.9)";
        ctx.shadowBlur = 10;
        ctx.shadowColor = "rgba(255,215,120,0.7)";
        ctx.fill();
        ctx.shadowBlur = 0;
    }
}

for (let i = 0; i < 60; i++) particles.push(new Particle());

function connect() {
    for (let a = 0; a < particles.length; a++) {
        for (let b = a + 1; b < particles.length; b++) {
            const dx = particles[a].x - particles[b].x;
            const dy = particles[a].y - particles[b].y;
            const dist = Math.sqrt(dx * dx + dy * dy);
            if (dist < 140) {
                const opacity = 0.35 * (1 - dist / 140);
                const glow = 1.5 - dist / 200;
                ctx.strokeStyle = `rgba(255,215,100,${opacity})`;
                ctx.lineWidth = 0.9;
                ctx.shadowBlur = 6 * glow;
                ctx.shadowColor = "rgba(255,215,120,0.9)";
                ctx.beginPath();
                ctx.moveTo(particles[a].x, particles[a].y);
                ctx.lineTo(particles[b].x, particles[b].y);
                ctx.stroke();
                ctx.shadowBlur = 0;
            }
        }
    }
}

function animate() {
    ctx.clearRect(0, 0, w, h);
    particles.forEach(p => {
        p.move();
        p.draw();
    });
```

```
        connect();
        requestAnimationFrame(animate);
    }
    animate();

    /* === Tips === */
    const tips = [
        "LEARN SOMETHING NEW EVERY DAY",
        "CURIOSITY CREATES INNOVATION",
        "PROGRESS BEATS PERFECTION",
        "THINK. CREATE. EVOLVE.",
        "STAY FOCUSED AND CONSISTENT",
        "EFFORT BUILDS EXCELLENCE"
    ];
    const tipText = document.getElementById("tip-text");
    const tipSection = document.getElementById("tips-section");
    let tipIndex = 0;

    function showNextTip() {
        tipText.style.opacity = 0;
        setTimeout(() => {
            tipText.textContent = tips[tipIndex];
            tipText.style.opacity = 1;
            tipIndex = (tipIndex + 1) % tips.length;
        }, 500);
    }
    showNextTip();
    setInterval(showNextTip, 6000);
  </script>

</body>

</html>

<!DOCTYPE html>
<html lang="en">


<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>A.C.E Login | TANIA</title>
<style>
    * {
        margin: 0;
        padding: 0;
        box-sizing: border-box;
        font-family: "Tahoma", sans-serif;
    }

    body {
        height: 100vh;
        overflow: hidden;
        display: flex;
        justify-content: center;
        align-items: center;
        background: #000;
        color: #d4af37;
        position: relative;
    }

    canvas#network {
        position: fixed;
        inset: 0;
        z-index: -1;
        background: radial-gradient(circle at top left, #0a0a0a 0%, #000 90%);
    }

    .container {
        width: 400px;
```

```
    padding: 35px 30px 25px;
    background: rgba(0, 0, 0, 0.85);
    border: 1px solid #d4af37;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 25px rgba(212, 175, 55, 0.15);
    z-index: 1;
}

.logo img {
    width: 140px;
    border-radius: 10px;
    margin-bottom: 15px;
}

.title {
    font-size: 1.6em;
    font-weight: 600;
    margin-bottom: 4px;
    color: #d4af37;
}

.byline {
    font-size: 0.9em;
    color: #bfa94f;
    margin-bottom: 12px;
    letter-spacing: 1px;
}

.role-buttons {
    display: flex;
```

```
    justify-content: space-between;
    margin-bottom: 15px;
}

.role-btn {
    flex: 1;
    background: none;
    border: 1px solid #d4af37;
    color: #d4af37;
    padding: 8px 0;
    border-radius: 6px;
    margin: 0 5px;
    cursor: pointer;
    font-size: 0.85em;
    transition: 0.3s;
}

.role-btn:hover,
.role-btn.active {
    background: #d4af37;
    color: #000;
}

#error-message {
    color: #ff4d4d;
    font-size: 0.9em;
    min-height: 20px;
    margin-bottom: 10px;
}

label {
```

```
    font-size: 0.9em;
    color: #bfa94f;
    margin: 8px 0 4px 3px;
    display: block;
    text-align: left;
}

input {
    width: 100%;
    background: #000;
    color: #fff;
    border: 1px solid #d4af37;
    border-radius: 6px;
    padding: 10px;
    font-size: 1em;
    margin-bottom: 10px;
}

button.login-btn {
    margin-top: 10px;
    background: #d4af37;
    color: #000;
    font-weight: bold;
    padding: 10px;
    border: none;
    border-radius: 6px;
    font-size: 1em;
    cursor: pointer;
    transition: .3s;
}
```

```
        button.login-btn:hover {
            background: #c2a12e;
        }

        .links {
            margin-top: 12px;
            text-align: center;
            font-size: 0.85em;
        }

        .links a {
            color: #b2962d;
            text-decoration: none;
            margin: 0 6px;
            transition: .3s;
        }

        .links a:hover {
            color: #fff;
        }

        footer {
            position: absolute;
            bottom: 12px;
            width: 100%;
            text-align: center;
            color: #777;
            font-size: 0.75em;
        }
    </style>
</head>
```

```
<body>
  <canvas id="network"></canvas>

  <div class="container">
    <div class="logo">
      <img src="https://i.postimg.cc/sfm33x2Q/Untitled-document-7-1-page-0001.jpg" alt="A.C.E
Logo">
    </div>
    <div class="title">A.C.E</div>
    <div class="byline">Aspire • Connect • Evolve</div>

    <div class="role-buttons">
      <button class="role-btn" onclick="selectRole(this)">Student</button>
      <button class="role-btn" onclick="selectRole(this)">Teacher</button>
      <button class="role-btn" onclick="selectRole(this)">Parent</button>
    </div>

    <div id="error-message"></div>

    <label for="username">Email</label>
    <input id="username" type="email" placeholder="Enter your email" required>
    <label for="password">Password</label>
    <input id="password" type="password" placeholder="Enter your password" required>

    <button class="login-btn" onclick="login()">Login</button>

    <div class="links">
      <a href="#">Forgot Password?</a> | <a href="#">Sign Up</a>
    </div>
  </div>
```

```
<footer>© 2025 A.C.E Initiative | TANIA Prototype</footer>

<script>
  let selectedRole = null;

  function selectRole(btn) {
    document.querySelectorAll('.role-btn').forEach(b => b.classList.remove('active'));
    btn.classList.add('active');
    selectedRole = btn.textContent.trim();
    document.getElementById('error-message').textContent = "";
  }

  async function login() {
    const email = document.getElementById('username').value.trim();
    const password = document.getElementById('password').value.trim();
    const errorEl = document.getElementById('error-message');

    if (!selectedRole) return errorEl.textContent = "   Please select your role.";
    if (!email || !password) return errorEl.textContent = "   Enter email and password.";

    try {
      const res = await fetch("http://localhost:5050/api/users/login", {
        method: "POST",
        headers: {
          "Content-Type": "application/json"
        },
        body: JSON.stringify({
          email,
          password,
          role: selectedRole
```

```
        })
      });
      const data = await res.json();
      if (data.success) {
        localStorage.setItem("userRole", selectedRole);
        localStorage.setItem("userName", data.user.name || email);
        errorEl.style.color = "#00cc66";
        errorEl.textContent = "✅ Login Successful!";
        setTimeout(() => {
          if (selectedRole === "Teacher") window.location.href = "teacher-dashboard.html";
          else if (selectedRole === "Student") window.location.href = "student-dashboard.html";
          else window.location.href = "parent-dashboard.html";
        }, 700);
      } else errorEl.textContent = data.message || "❌ Invalid credentials.";
    } catch (e) {
      errorEl.textContent = "⚠ Server unreachable.";
    }
}

/* Background gold particle animation */
const c = document.getElementById("network"),
  x = c.getContext("2d");
let W, H, P = [];

function resize() {
  W = c.width = innerWidth;
  H = c.height = innerHeight;
}
window.addEventListener("resize", resize);
resize();
class Particle {
```

```
  constructor() {
    this.x = Math.random() * W;
    this.y = Math.random() * H;
    this.vx = (Math.random() - .5) * .5;
    this.vy = (Math.random() - .5) * .5;
  }
  move() {
    this.x += this.vx;
    this.y += this.vy;
    if (this.x < 0 || this.x > W) this.vx *= -1;
    if (this.y < 0 || this.y > H) this.vy *= -1;
  }
  draw() {
    x.beginPath();
    x.arc(this.x, this.y, 1.4, 0, 2 * Math.PI);
    x.fillStyle = "rgba(255,215,120,.9)";
    x.shadowBlur = 10;
    x.shadowColor = "gold";
    x.fill();
    x.shadowBlur = 0;
  }
}
for (let i = 0; i < 65; i++) P.push(new Particle());

function connect() {
  for (let a = 0; a < P.length; a++) {
    for (let b = a + 1; b < P.length; b++) {
      const dx = P[a].x - P[b].x,
        dy = P[a].y - P[b].y,
        d = Math.sqrt(dx * dx + dy * dy);
      if (d < 140) {
```

```
                x.strokeStyle = `rgba(255,215,120,${0.3*(1-d/140)})`;
                x.beginPath();
                x.moveTo(P[a].x, P[a].y);
                x.lineTo(P[b].x, P[b].y);
                x.stroke();
            }
        }
    }
}
(function anim() {
    x.clearRect(0, 0, W, H);
    P.forEach(p => {
        p.move();
        p.draw();
    });
    connect();
    requestAnimationFrame(anim)
})();
if (data.success) {
    localStorage.setItem("userRole", selectedRole);
    localStorage.setItem("userName", data.user.name || email);

    errorEl.style.color = "#00cc66";
    errorEl.textContent = "✅ Login Successful!";

    setTimeout(() => {
        if (selectedRole === "Teacher") {
            window.location.href = "teacher-portal.html"; //    now goes to portal
        } else if (selectedRole === "Student") {
            window.location.href = "student-dashboard.html";
        } else {
```

```
                window.location.href = "parent-dashboard.html";
            }
        }, 700);
    }
  </script>
</body>

</html>

const canvas = document.getElementById("network");
if (canvas) {
  const ctx = canvas.getContext("2d");
  let w, h, particles = [];

  function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
  }
  window.addEventListener("resize", resize);
  resize();

  class Particle {
    constructor() {
      this.x = Math.random() * w;
      this.y = Math.random() * h;
      this.vx = (Math.random() - 0.5) * 1.2;
      this.vy = (Math.random() - 0.5) * 1.2;
      this.size = 1.6;
    }
    move() {
      this.x += this.vx;
```

```
        this.y += this.vy;
        if (this.x < 0 || this.x > w) this.vx *= -1;
        if (this.y < 0 || this.y > h) this.vy *= -1;
    }
    draw() {
        ctx.beginPath();
        ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
        ctx.fillStyle = "rgba(255,215,120,0.9)";
        ctx.shadowBlur = 12;
        ctx.shadowColor = "rgba(255,215,120,0.9)";
        ctx.fill();
        ctx.shadowBlur = 0;
    }
}

for (let i = 0; i < 80; i++) particles.push(new Particle());

function connect() {
    for (let a = 0; a < particles.length; a++) {
        for (let b = a + 1; b < particles.length; b++) {
            const dx = particles[a].x - particles[b].x;
            const dy = particles[a].y - particles[b].y;
            const dist = Math.sqrt(dx * dx + dy * dy);
            if (dist < 140) {
                const opacity = 0.35 * (1 - dist / 140);
                ctx.strokeStyle = `rgba(255,215,100,${opacity})`;
                ctx.lineWidth = 0.9;
                ctx.shadowBlur = 6;
                ctx.shadowColor = "rgba(255,215,120,0.9)";
                ctx.beginPath();
                ctx.moveTo(particles[a].x, particles[a].y);
```

```
                ctx.lineTo(particles[b].x, particles[b].y);
                ctx.stroke();
                ctx.shadowBlur = 0;
            }
        }
    }
  }

  function animate() {
    ctx.fillStyle = "rgba(0,0,0,0.2)";
    ctx.fillRect(0, 0, w, h);
    particles.forEach(p => { p.move();
      p.draw(); });
    connect();
    requestAnimationFrame(animate);
  }
  animate();
}

const canvas = document.getElementById("network");
if (canvas) {
  const ctx = canvas.getContext("2d");
  let w, h, particles = [];

  function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
  }
  window.addEventListener("resize", resize);
  resize();
```

```
class Particle {
  constructor() {
    this.x = Math.random() * w;
    this.y = Math.random() * h;
    this.vx = (Math.random() - 0.5) * 1.2;
    this.vy = (Math.random() - 0.5) * 1.2;
    this.size = 1.6;
  }
  move() {
    this.x += this.vx;
    this.y += this.vy;
    if (this.x < 0 || this.x > w) this.vx *= -1;
    if (this.y < 0 || this.y > h) this.vy *= -1;
  }
  draw() {
    ctx.beginPath();
    ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
    ctx.fillStyle = "rgba(255,215,120,0.9)";
    ctx.shadowBlur = 12;
    ctx.shadowColor = "rgba(255,215,120,0.9)";
    ctx.fill();
    ctx.shadowBlur = 0;
  }
}

for (let i = 0; i < 80; i++) particles.push(new Particle());

function connect() {
  for (let a = 0; a < particles.length; a++) {
    for (let b = a + 1; b < particles.length; b++) {
      const dx = particles[a].x - particles[b].x;
```

```
            const dy = particles[a].y - particles[b].y;
            const dist = Math.sqrt(dx * dx + dy * dy);
            if (dist < 140) {
              const opacity = 0.35 * (1 - dist / 140);
              ctx.strokeStyle = `rgba(255,215,100,${opacity})`;
              ctx.lineWidth = 0.9;
              ctx.shadowBlur = 6;
              ctx.shadowColor = "rgba(255,215,120,0.9)";
              ctx.beginPath();
              ctx.moveTo(particles[a].x, particles[a].y);
              ctx.lineTo(particles[b].x, particles[b].y);
              ctx.stroke();
              ctx.shadowBlur = 0;
            }
          }
        }
      }

      function animate() {
        ctx.fillStyle = "rgba(0,0,0,0.2)";
        ctx.fillRect(0, 0, w, h);
        particles.forEach(p => { p.move();
          p.draw(); });
        connect();
        requestAnimationFrame(animate);
      }
      animate();
    }

    <!DOCTYPE html>
    <html lang="en">
```

```
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>A.C.E | RIZA — Parent Portal</title>

  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: "Georgia", serif;
    }

    body {
      height: 100vh;
      overflow: hidden;
      background: radial-gradient(circle at top left, #0a0a0a 0%, #000 90%);
      color: #d4af37;
      display: flex;
      justify-content: center;
      align-items: center;
      position: relative;
    }

    canvas#network {
      position: fixed;
      inset: 0;
      z-index: -1;
    }
```

```
header {
    position: fixed;
    top: 80px;
    width: 100%;
    text-align: center;
    z-index: 5;
    transition: all .7s ease;
}

.header-logo img {
    width: 220px;
    border-radius: 14px;
    border: 2px solid #d4af37;
    box-shadow: 0 0 35px rgba(212, 175, 55, 0.35);
    transition: all .7s ease;
}

.header-logo.small img {
    width: 90px;
    box-shadow: 0 0 15px rgba(212, 175, 55, 0.25);
}

.header-title {
    font-size: 2.8em;
    font-weight: bold;
    color: #ffdb58;
    letter-spacing: 2px;
    text-transform: uppercase;
    margin-top: 20px;
    opacity: 1;
    transition: opacity .5s ease;
```

```
}

.header-title.hidden {
    opacity: 0;
}

.header-tagline {
    font-size: 1.05em;
    color: #bfa94f;
    letter-spacing: 3px;
    text-transform: uppercase;
    margin-top: 6px;
    opacity: 1;
    transition: opacity .5s ease;
}

.header-tagline.hidden {
    opacity: 0;
}

.status-bar {
    position: fixed;
    top: 20px;
    right: 30px;
    display: flex;
    flex-direction: column;
    align-items: flex-end;
    background: rgba(0, 0, 0, 0.55);
    border: 1px solid rgba(212, 175, 55, 0.4);
    padding: 10px 18px;
    border-radius: 10px;
```

```
    font-size: 1em;
    box-shadow: 0 0 12px rgba(212, 175, 55, 0.1);
    backdrop-filter: blur(6px);
    z-index: 6;
}

.status-bar div {
    color: #ffdb58;
    font-weight: 600;
    letter-spacing: 1px;
    margin: 2px 0;
}

.menu-icon {
    position: fixed;
    top: 25px;
    left: 25px;
    font-size: 28px;
    color: #d4af37;
    cursor: pointer;
    z-index: 10;
    transition: .3s;
}

.menu-icon:hover {
    color: #ffdb58;
    transform: scale(1.1);
}

.sidebar {
    position: fixed;
```

```
    top: 0;
    left: -250px;
    width: 250px;
    height: 100%;
    background: rgba(0, 0, 0, .96);
    border-right: 1px solid #d4af37;
    padding-top: 120px;
    transition: left .4s ease;
    z-index: 9;
}

.sidebar.active {
    left: 0;
}

.sidebar ul {
    list-style: none;
}

.sidebar ul li {
    padding: 16px 30px;
    font-size: 1em;
    color: #d4af37;
    cursor: pointer;
    transition: .3s;
    border-bottom: 1px solid rgba(212, 175, 55, .15);
    position: relative;
}

.sidebar ul li:hover,
.sidebar ul li.active {
```

```
    background: rgba(212, 175, 55, .15);
    color: #ffdb58;
    padding-left: 45px;
}
/* === RIZA Tooltip === */

.riza-tooltip {
    position: relative;
    cursor: help;
    color: #ffdb58;
    font-weight: bold;
}

.riza-tooltip::after {
    content: "Responsive Intelligent Zenith Advisor";
    position: absolute;
    bottom: 125%;
    left: 50%;
    transform: translateX(-50%);
    background: rgba(0, 0, 0, 0.9);
    color: #ffdb58;
    padding: 6px 12px;
    border: 1px solid #d4af37;
    border-radius: 6px;
    white-space: nowrap;
    font-size: 0.8em;
    opacity: 0;
    pointer-events: none;
    transition: opacity .25s ease;
    box-shadow: 0 0 12px rgba(212, 175, 55, 0.15);
}
```

```
.riza-tooltip:hover::after {
    opacity: 1;
}

.container {
    background: rgba(0, 0, 0, .82);
    border: 1px solid #d4af37;
    border-radius: 15px;
    width: 750px;
    padding: 40px 60px;
    box-shadow: 0 0 25px rgba(212, 175, 55, .15);
    text-align: center;
    position: relative;
    z-index: 1;
    margin-top: 300px;
    transition: all .4s ease;
}

.content h2 {
    font-size: 1.5em;
    color: #ffdb58;
    margin-bottom: 15px;
    border-bottom: 1px solid rgba(212, 175, 55, .3);
    padding-bottom: 5px;
}

.info {
    line-height: 1.8em;
    font-size: .95em;
    color: #bfa94f;
```

```
    text-align: left;
}

.info span {
    color: #d4af37;
    font-weight: bold;
}

footer {
    position: fixed;
    bottom: 12px;
    width: 100%;
    text-align: center;
    color: #777;
    font-size: .8em;
    z-index: 6;
}
/* === RIZA Chat Button === */

.assistant-btn {
    position: fixed;
    bottom: 25px;
    right: 30px;
    width: 65px;
    height: 65px;
    background: radial-gradient(circle, #000 40%, #d4af37 180%);
    border: 2px solid #d4af37;
    border-radius: 50%;
    display: flex;
    justify-content: center;
    align-items: center;
```

```
    cursor: pointer;
    box-shadow: 0 0 20px rgba(212, 175, 55, 0.4);
    transition: .3s;
    z-index: 8;
}

.assistant-btn:hover {
    transform: scale(1.12);
    box-shadow: 0 0 30px rgba(255, 215, 100, 0.6);
}

.assistant-btn svg {
    width: 34px;
    height: 34px;
    fill: #ffdb58;
}

.chatbox {
    position: fixed;
    bottom: 100px;
    right: 40px;
    width: 320px;
    height: 400px;
    background: rgba(0, 0, 0, .9);
    border: 1px solid #d4af37;
    border-radius: 15px;
    display: none;
    flex-direction: column;
    overflow: hidden;
    box-shadow: 0 0 25px rgba(212, 175, 55, .3);
    z-index: 12;
```

```
}

.chat-header {
    background: rgba(212, 175, 55, .1);
    padding: 10px;
    color: #ffdb58;
    font-weight: bold;
    border-bottom: 1px solid #d4af37;
    text-align: center;
}

.chat-messages {
    flex: 1;
    padding: 10px;
    overflow-y: auto;
    color: #d4af37;
    font-size: 0.9em;
}

.chat-input {
    display: flex;
    border-top: 1px solid #d4af37;
}

.chat-input input {
    flex: 1;
    padding: 8px;
    background: #000;
    border: none;
    color: #ffdb58;
    outline: none;
```

```
}

.chat-input button {
    background: #d4af37;
    color: #000;
    border: none;
    padding: 0 12px;
    cursor: pointer;
    font-weight: bold;
}

.message {
    margin: 6px 0;
}

.user {
    color: #ffdb58;
    text-align: right;
}

.bot {
    color: #d4af37;
    text-align: left;
    background: rgba(255, 215, 50, 0.05);
    padding: 6px 8px;
    border-radius: 6px;
    border-left: 2px solid #ffdb58;
}
/* Guided Tour */

#guide-overlay {
```

```
    position: fixed;
    inset: 0;
    background: rgba(0, 0, 0, 0.9);
    color: #ffdb58;
    display: none;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    text-align: center;
    z-index: 1000;
    animation: fadeIn .25s ease;
}

#guide-box {
    border: 1px solid #d4af37;
    background: rgba(0, 0, 0, 0.85);
    padding: 40px 60px;
    border-radius: 15px;
    width: 520px;
    max-width: 92vw;
    box-shadow: 0 0 25px rgba(212, 175, 55, 0.3);
}

#guide-box h2 {
    margin-bottom: 10px;
}

#guide-text {
    min-height: 84px;
    color: #d4af37;
    line-height: 1.6;
```

```
    margin: 10px 0 18px;
}

#guide-next {
    background: #d4af37;
    color: #000;
    padding: 10px 18px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    font-weight: bold;
    transition: .2s;
    min-width: 110px;
}

#guide-next[disabled] {
    opacity: .5;
    cursor: not-allowed;
}

.cursor {
    display: inline-block;
    margin-left: 4px;
    opacity: 1;
    animation: blink 1s step-end infinite;
}

@keyframes blink {
    50% {
        opacity: 0;
    }
```

```
}

@keyframes fadeIn {
    from {
        opacity: 0
    }
    to {
        opacity: 1
    }
}
/* Logout Overlay */

#logout-overlay {
    position: fixed;
    inset: 0;
    background: rgba(0, 0, 0, 0.8);
    display: none;
    justify-content: center;
    align-items: center;
    z-index: 999;
}

#logout-box {
    background: rgba(0, 0, 0, 0.9);
    border: 1px solid #d4af37;
    border-radius: 12px;
    padding: 28px 34px;
    text-align: center;
    color: #ffdb58;
    box-shadow: 0 0 25px rgba(212, 175, 55, .3);
}
```

```
        #logout-box button {
            margin: 10px 8px;
            padding: 8px 16px;
            border-radius: 6px;
            font-weight: bold;
            cursor: pointer;
        }

        #logout-yes {
            background: #d4af37;
            color: #000;
            border: none;
        }

        #logout-no {
            background: transparent;
            color: #ffdb58;
            border: 1px solid #d4af37;
        }
    </style>
</head>

<body>
    <canvas id="network"></canvas>

    <header id="header">
        <div class="header-logo" id="headerLogo">
            <img src="https://i.postimg.cc/xjy79m1t/Whats-App-Image-2025-10-12-at-13-32-07-
54be146c.jpg" alt="A.C.E Logo">
        </div>
```

```
    <div class="header-title" id="headerTitle">Welcome to A.C.E</div>
    <div class="header-tagline" id="headerTagline">Advise • Care • Empower</div>
</header>

<div class="status-bar">
    <div>Role: Parent</div>
    <div id="datetime">--:--:-- | -- - -------- - ----</div>
</div>

<div class="menu-icon" onclick="toggleMenu()">☰</div>

<nav class="sidebar" id="sidebar">
    <ul>
        <li onclick="showSection(this,'home')">   Home</li>
        <li onclick="showSection(this,'child')">       Child Overview</li>
        <li onclick="showSection(this,'dashboard')">   Student Dashboard</li>
        <li onclick="showSection(this,'academic')">   Academic Skills</li>
        <li onclick="showSection(this,'career')">   Career Suggestions</li>
        <li class="riza-tooltip" onclick="showSection(this,'riza')">   RIZA</li>
        <li onclick="showSection(this,'notes')">   Teacher Notes</li>
        <li onclick="showSection(this,'todo')">   To-Do List</li>
        <li onclick="showSection(this,'faq')">❓ FAQ</li>
        <li onclick="showSection(this,'support')">   Support</li>
        <li onclick="showLogout()">   Logout</li>
    </ul>
</nav>

<div class="container" id="content-area">
    <div id="content" class="content"></div>
</div>
```

```
<div class="assistant-btn" onclick="toggleChat()" title="Chat with RIZA">
    <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24"><path d="M12 2a10 10 0 0 0 0-
10 10c0 5.52 4.48 10 10 10 2.39 0 4.6-.84 6.34-2.25L22 22v-4.66A9.93 9.93 0 0 0 22 12 10 10 0 0 0 12
2zm-1 14h2v2h-2v-2zm1-12a8 8 0 0 1 8 8c0 1.85-.63 3.55-1.69 4.9L18 17l-.31-.1C17.16 16.64 17
15.85 17 15H7v-2h10v-2H7V9h8V7H7V5h5z"/></svg>
</div>

<div class="chatbox" id="chatbox">
    <div class="chat-header riza-tooltip">   RIZA — Responsive Intelligent Zenith Advisor</div>
    <div class="chat-messages" id="chat-messages">
        <div class="message bot"><strong>   RIZA:</strong> Hello! I'm RIZA, your A.C.E guide. How
can I assist you today?</div>
    </div>
    <div class="chat-input">
        <input type="text" id="user-input" placeholder="Ask RIZA...">
        <button onclick="sendMessage()">Send</button>
    </div>
</div>

<!-- Logout Overlay -->
<div id="logout-overlay">
    <div id="logout-box">
        <h3>   Confirm Logout</h3>
        <p>Are you sure you want to log out?</p>
        <button id="logout-yes">Yes</button>
        <button id="logout-no">No</button>
    </div>
</div>

<!-- Guided Walkthrough -->
<div id="guide-overlay" aria-hidden="true">
    <div id="guide-box">
```

```
        <h2>Welcome to A.C.E</h2>
        <p id="guide-text"></p>
        <button id="guide-next" disabled>Next</button>
      </div>
    </div>

    <footer>© 2025 A.C.E Parent Portal | Powered by <span class="riza-tooltip">
RIZA</span></footer>

    <script>
      const content = document.getElementById("content");
      const headerLogo = document.getElementById("headerLogo");
      const headerTitle = document.getElementById("headerTitle");
      const headerTagline = document.getElementById("headerTagline");

      const sections = {
        home: `<h2>Welcome, Mr. & Mrs. Sharma  </h2><div class='info'><p>Welcome to your A.C.E
Parent Portal powered by <span class='riza-tooltip'>  RIZA</span>.</p><p>Here you can monitor
Aarav's progress, skills, and teacher feedback in one connected ecosystem.</p></div>`,
        child: `<h2>Child Overview</h2><div class='info'><p><span>Name:</span> Aarav
Sharma</p><p><span>Grade:</span> BCA 2nd Year</p><p><span>Roll No:</span>
23BCA0101</p></div>`,
        dashboard: `<h2>  Student Dashboard</h2><div class='info'><p><span>Attendance:</span>
92%</p><p><span>Assignments:</span> 8 / 10</p></div>`,
        academic: `<h2>Academic Skills</h2><div class='info'><p><span>Subjects:</span> Math,
Physics, CS</p><p><span>Skills:</span> Python, ML, Data Analytics</p></div>`,
        career: `<h2>Career Suggestions</h2><div class='info'><ul><li>AI Engineer</li><li>Data
Scientist</li><li>Software Developer</li></ul></div>`,
        notes: `<h2>Teacher Notes</h2><div class='info'><p>"Aarav is showing consistent progress in
Machine Learning."</p></div>`,
        todo: `<h2>To-Do List</h2><ul class='info'><li>Submit AI project</li><li>Prepare for
test</li></ul>`,
        faq: `<h2> ❓ FAQ</h2><div class='info'><p><strong>Q:</strong> How can I track my child's
progress?<br><strong>A:</strong> Visit the Student Dashboard
section.</p><p><strong>Q:</strong> What is RIZA?<br><strong>A:</strong> <span class='riza-
```

```
tooltip'>RIZA</span> is your Responsive Intelligent Zenith Advisor that helps analyze and suggest improvements for your child.</p></div>`,
        support: `<h2>Support</h2><div class='info'><p>   1800-ACE-HELP</p><p> parent@ace.ai</p></div>`
    };

    function toggleMenu() {
        document.getElementById("sidebar").classList.toggle("active");
    }

    function showSection(el, section) {
        document.querySelectorAll(".sidebar ul li").forEach(li => li.classList.remove("active"));
        el.classList.add("active");
        if (section === "riza") {
            toggleChat();
            return;
        }
        content.innerHTML = sections[section];
        if (section !== "home") {
            headerLogo.classList.add("small");
            headerTitle.classList.add("hidden");
            headerTagline.classList.add("hidden");
        } else {
            headerLogo.classList.remove("small");
            headerTitle.classList.remove("hidden");
            headerTagline.classList.remove("hidden");
        }
    }

    function toggleChat() {
        const chat = document.getElementById("chatbox");
        chat.style.display = (chat.style.display === "flex") ? "none" : "flex";
```

```
}

function sendMessage() {
  const input = document.getElementById("user-input");
  const msg = input.value.trim();
  if (!msg) return;
  const chat = document.getElementById("chat-messages");
  chat.innerHTML += `<div class='message user'>${msg}</div>`;
  input.value = "";
  setTimeout(() => {
    chat.innerHTML += `<div class='message bot'><strong>   RIZA:</strong> Got it! Let’s work
on that together.</div>`;
    chat.scrollTop = chat.scrollHeight;
  }, 700);
}

/* Logout Function */
function showLogout() {
  const overlay = document.getElementById("logout-overlay");
  overlay.style.display = "flex";
  document.getElementById("logout-yes").onclick = () => {
    overlay.style.display = "none";
    alert("You’ve been logged out successfully!");
    window.location.href = "login.html";
  };
  document.getElementById("logout-no").onclick = () => {
    overlay.style.display = "none";
  };
}

/* Clock */
```

```
function updateDateTime() {
    const now = new Date();
    document.getElementById('datetime').textContent = `${now.toLocaleTimeString('en-GB',{hour12:false})} | ${now.toLocaleDateString('en-GB',{day:'numeric',month:'long',year:'numeric'})}`;
}
setInterval(updateDateTime, 1000);
updateDateTime();

/* Background Animation */
const canvas = document.getElementById("network"),
    ctx = canvas.getContext("2d");
let w, h, particles = [];

function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
}
window.addEventListener("resize", resize);
resize();
class Particle {
    constructor() {
        this.x = Math.random() * w;
        this.y = Math.random() * h;
        this.vx = (Math.random() - .5) * 1.1;
        this.vy = (Math.random() - .5) * 1.1;
        this.size = 1.6;
    }
    move() {
        this.x += this.vx;
        this.y += this.vy;
        if (this.x < 0 || this.x > w) this.vx *= -1;
```

```
        if (this.y < 0 || this.y > h) this.vy *= -1;
    }
    draw() {
        ctx.beginPath();
        ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
        ctx.fillStyle = "rgba(255,215,120,0.9)";
        ctx.shadowBlur = 6;
        ctx.shadowColor = "rgba(255,215,100,0.9)";
        ctx.fill();
        ctx.shadowBlur = 0;
    }
}
for (let i = 0; i < 80; i++) particles.push(new Particle());

function connect() {
    for (let a = 0; a < particles.length; a++) {
        for (let b = a + 1; b < a + 30 && b < particles.length; b++) {
            const dx = particles[a].x - particles[b].x,
                dy = particles[a].y - particles[b].y;
            const dist = Math.sqrt(dx * dx + dy * dy);
            if (dist < 130) {
                const op = 0.3 * (1 - dist / 130);
                ctx.strokeStyle = `rgba(255,215,120,${op})`;
                ctx.lineWidth = 0.8;
                ctx.shadowBlur = 4;
                ctx.shadowColor = "rgba(255,215,100,0.9)";
                ctx.beginPath();
                ctx.moveTo(particles[a].x, particles[a].y);
                ctx.lineTo(particles[b].x, particles[b].y);
                ctx.stroke();
                ctx.shadowBlur = 0;
```

```
            }
        }
    }
}

function animate() {
    ctx.clearRect(0, 0, w, h);
    particles.forEach(p => {
        p.move();
        p.draw();
    });
    connect();
    requestAnimationFrame(animate);
}
animate();

/* Walkthrough */
const guideOverlay = document.getElementById('guide-overlay');
const guideText = document.getElementById('guide-text');
const guideNext = document.getElementById('guide-next');
const guideSteps = [
    "   Hello! I'm <span class='riza-tooltip'>   RIZA</span> — your Responsive Intelligent Zenith Advisor.",
    "Welcome to the A.C.E Parent Portal, designed to help you Advise • Care • Empower.",
    "   Student Dashboard — track attendance and overall academic progress.",
    "   Academic Skills — review your child’s subject strengths and growth areas.",
    "   Career Suggestions — explore data-driven career insights.",
    "   Teacher Notes — stay updated with instructor feedback.",
    "   To-Do List — track upcoming submissions and deadlines.",
    " ?  FAQ — find quick answers and support details.",
    "That’s all! Let’s help your child reach their zenith together   "
```

```
];
let guideIndex = 0;

function typeStep(text, cb) {
  guideText.innerHTML = "";
  guideNext.disabled = true;
  const cursor = document.createElement('span');
  cursor.className = 'cursor';
  cursor.textContent = '|';
  guideText.appendChild(cursor);
  let i = 0;
  const interval = setInterval(() => {
    cursor.remove();
    guideText.innerHTML = text.slice(0, i + 1);
    guideText.appendChild(cursor);
    i++;
    if (i >= text.length) {
      clearInterval(interval);
      cursor.remove();
      guideNext.disabled = false;
      if (cb) cb();
    }
  }, 35);
}

function startTour() {
  guideOverlay.style.display = 'flex';
  guideOverlay.setAttribute('aria-hidden', 'false');
  guideIndex = 0;
  typeStep(guideSteps[guideIndex]);
  guideNext.onclick = () => {
```

```
            guideIndex++;
            if (guideIndex < guideSteps.length) {
                typeStep(guideSteps[guideIndex]);
            } else {
                endTour();
            }
        };
    }

    function endTour() {
        guideOverlay.style.display = 'none';
        localStorage.setItem('ace_parent_guide', 'true');
    }
    if (!localStorage.getItem('ace_parent_guide')) {
        setTimeout(startTour, 800);
    }

    /* Load Home Section Initially */
    content.innerHTML = sections.home;
  </script>
</body>

</html>

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>A.C.E Register | Tap for Tips</title>
```

```
<style>
  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: "Tahoma", sans-serif;
  }

  body {
    height: 100vh;
    overflow: hidden;
    display: flex;
    justify-content: center;
    align-items: center;
    background: #000;
    color: #d4af37;
    position: relative;
  }

  canvas#network {
    position: fixed;
    inset: 0;
    z-index: -1;
    background: radial-gradient(circle at top left, #0a0a0a 0%, #000 90%);
  }

  .container {
    width: 400px;
    padding: 35px 30px 25px;
    background: rgba(0, 0, 0, 0.85);
    border: 1px solid #d4af37;
```

```
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 25px rgba(212, 175, 55, 0.15);
    z-index: 1;
}

.logo img {
    width: 140px;
    border-radius: 10px;
    margin-bottom: 15px;
}

.title {
    font-size: 1.6em;
    font-weight: 600;
    margin-bottom: 4px;
    color: #d4af37;
}

.byline {
    font-size: 0.9em;
    color: #bfa94f;
    margin-bottom: 12px;
    letter-spacing: 1px;
}

#error-message {
    color: #ff4d4d;
    font-size: 0.9em;
    min-height: 20px;
    margin-bottom: 10px;
```

```
}

form {
    text-align: left;
    display: flex;
    flex-direction: column;
}

label {
    font-size: 0.9em;
    color: #bfa94f;
    margin: 8px 0 4px 3px;
}

input,
select {
    width: 100%;
    background: #000;
    color: #fff;
    border: 1px solid #d4af37;
    border-radius: 6px;
    padding: 10px;
    font-size: 1em;
    outline: none;
    transition: 0.3s;
    caret-color: #d4af37;
}

input:focus,
select:focus {
    border-color: #b9972e;
```

```
}

.login-btn {
    margin-top: 15px;
    background: #d4af37;
    color: #000;
    font-weight: bold;
    padding: 10px;
    border: none;
    border-radius: 6px;
    font-size: 1em;
    cursor: pointer;
    transition: .3s;
}

.login-btn:hover {
    background: #c2a12e;
}

.links {
    margin-top: 12px;
    text-align: center;
    font-size: 0.85em;
}

.links a {
    color: #b2962d;
    text-decoration: none;
    margin: 0 6px;
    transition: .3s;
}
```

```
.links a:hover {
    color: #fff;
}

.tips-section {
    margin-top: 18px;
    padding-top: 12px;
    border-top: 1px solid rgba(212, 175, 55, 0.3);
    color: #d4af37;
    text-align: center;
    font-size: 0.9rem;
    letter-spacing: 1px;
    text-transform: uppercase;
    cursor: pointer;
    user-select: none;
}

.tips-section span {
    display: block;
    font-size: 0.85rem;
    margin-top: 6px;
    color: #ffdb58;
    opacity: 0;
    transition: opacity 1.2s ease;
    font-weight: 600;
}

.tips-section:hover strong {
    color: #ffdb58;
    transition: color 0.3s ease;
```

```
        }

        footer {
            position: absolute;
            bottom: 12px;
            width: 100%;
            text-align: center;
            color: #777;
            font-size: 0.75em;
            letter-spacing: 0.5px;
        }
    </style>
</head>

<body>
    <canvas id="network"></canvas>

    <div class="container">
        <div class="logo">
            <img src="https://i.postimg.cc/xjy79m1t/Whats-App-Image-2025-10-12-at-13-32-07-
54be146c.jpg" alt="A.C.E Logo">
        </div>
        <div class="title">Create Account</div>
        <div class="byline">Join the A.C.E Journey</div>

        <div id="error-message"></div>

        <form id="registerForm">
            <label for="name">Full Name</label>
            <input id="name" type="text" placeholder="Enter your name" required>
```

```
      <label for="email">Email</label>
      <input id="email" type="email" placeholder="Enter your email" required>

      <label for="password">Password</label>
      <input id="password" type="password" placeholder="Create a password" required>

      <label for="role">Select Role</label>
      <select id="role" required>
    <option value="">Select...</option>
    <option value="Student">Student</option>
    <option value="Teacher">Teacher</option>
    <option value="Parent">Parent</option>
  </select>

      <button type="submit" class="login-btn">Sign Up</button>
    </form>

    <div class="links">
      Already have an account? <a href="index.html">Login</a>
    </div>

    <div class="tips-section" id="tips-section">
      <strong>TIPS:</strong>
      <span id="tip-text"></span>
    </div>
  </div>

  <footer>© 2025 A.C.E Initiative | Concept Prototype</footer>

  <script>
    // === Registration Submit ===
```

```
const form = document.getElementById("registerForm");
const msg = document.getElementById("error-message");

form.addEventListener("submit", async(e) => {
  e.preventDefault();
  msg.textContent = "⏳ Registering...";
  msg.style.color = "#ffdb58";

  const name = document.getElementById("name").value.trim();
  const email = document.getElementById("email").value.trim();
  const password = document.getElementById("password").value.trim();
  const role = document.getElementById("role").value;

  try {
    const res = await fetch("/api/auth/register", {
      method: "POST",
      headers: {
        "Content-Type": "application/json"
      },
      body: JSON.stringify({
        name,
        email,
        password,
        role
      }),
    });
    const data = await res.json();

    if (res.ok) {
      msg.style.color = "#00cc66";
      msg.textContent = "✅ Registration successful!";
```

```
            setTimeout(() => (window.location.href = "index.html"), 1500);
        } else {
            msg.style.color = "#ff4d4d";
            msg.textContent = data.message || "❌ Registration failed";
        }
    } catch (err) {
        msg.style.color = "#ff4d4d";
        msg.textContent = "❌ Server error. Please try again.";
    }
});

// === Background Network (same as index.html) ===
const canvas = document.getElementById("network");
const ctx = canvas.getContext("2d");
let w, h, particles = [];

function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
}
window.addEventListener("resize", resize);
resize();

class Particle {
    constructor() {
        this.x = Math.random() * w;
        this.y = Math.random() * h;
        this.vx = (Math.random() - 0.5) * 0.3;
        this.vy = (Math.random() - 0.5) * 0.3;
        this.size = 1.5;
    }
```

```
  move() {
    this.x += this.vx;
    this.y += this.vy;
    if (this.x < 0 || this.x > w) this.vx *= -1;
    if (this.y < 0 || this.y > h) this.vy *= -1;
  }
  draw() {
    ctx.beginPath();
    ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
    ctx.fillStyle = "rgba(255,215,120,0.9)";
    ctx.shadowBlur = 10;
    ctx.shadowColor = "rgba(255,215,120,0.7)";
    ctx.fill();
    ctx.shadowBlur = 0;
  }
}

for (let i = 0; i < 60; i++) particles.push(new Particle());

function connect() {
  for (let a = 0; a < particles.length; a++) {
    for (let b = a + 1; b < particles.length; b++) {
      const dx = particles[a].x - particles[b].x;
      const dy = particles[a].y - particles[b].y;
      const dist = Math.sqrt(dx * dx + dy * dy);
      if (dist < 140) {
        const opacity = 0.35 * (1 - dist / 140);
        ctx.strokeStyle = `rgba(255,215,100,${opacity})`;
        ctx.lineWidth = 0.9;
        ctx.shadowBlur = 6;
        ctx.shadowColor = "rgba(255,215,120,0.9)";
```

```
            ctx.beginPath();
            ctx.moveTo(particles[a].x, particles[a].y);
            ctx.lineTo(particles[b].x, particles[b].y);
            ctx.stroke();
            ctx.shadowBlur = 0;
          }
        }
    }
}

function animate() {
    ctx.clearRect(0, 0, w, h);
    particles.forEach(p => {
        p.move();
        p.draw();
    });
    connect();
    requestAnimationFrame(animate);
}
animate();

// === Tips ===
const tips = [
    "LEARN SOMETHING NEW EVERY DAY",
    "CURIOSITY CREATES INNOVATION",
    "PROGRESS BEATS PERFECTION",
    "THINK. CREATE. EVOLVE.",
    "STAY FOCUSED AND CONSISTENT",
    "EFFORT BUILDS EXCELLENCE"
];
const tipText = document.getElementById("tip-text");
```

```
        let tipIndex = 0;

        function showNextTip() {
            tipText.style.opacity = 0;
            setTimeout(() => {
                tipText.textContent = tips[tipIndex];
                tipText.style.opacity = 1;
                tipIndex = (tipIndex + 1) % tips.length;
            }, 500);
        }
        showNextTip();
        setInterval(showNextTip, 6000);
    </script>
</body>

</html>

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>A.C.E Register | Tap for Tips</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: "Tahoma", sans-serif;
        }
```

```
body {
    height: 100vh;
    overflow: hidden;
    display: flex;
    justify-content: center;
    align-items: center;
    background: #000;
    color: #d4af37;
    position: relative;
}

canvas#network {
    position: fixed;
    inset: 0;
    z-index: -1;
    background: radial-gradient(circle at top left, #0a0a0a 0%, #000 90%);
}

.container {
    width: 400px;
    padding: 35px 30px 25px;
    background: rgba(0, 0, 0, 0.85);
    border: 1px solid #d4af37;
    border-radius: 10px;
    text-align: center;
    box-shadow: 0 0 25px rgba(212, 175, 55, 0.15);
    z-index: 1;
}

.logo img {
```

```
    width: 140px;
    border-radius: 10px;
    margin-bottom: 15px;
}

.title {
    font-size: 1.6em;
    font-weight: 600;
    margin-bottom: 4px;
    color: #d4af37;
}

.byline {
    font-size: 0.9em;
    color: #bfa94f;
    margin-bottom: 12px;
    letter-spacing: 1px;
}

#error-message {
    color: #ff4d4d;
    font-size: 0.9em;
    min-height: 20px;
    margin-bottom: 10px;
}

form {
    text-align: left;
    display: flex;
    flex-direction: column;
}
```

```
label {
    font-size: 0.9em;
    color: #bfa94f;
    margin: 8px 0 4px 3px;
}

input,
select {
    width: 100%;
    background: #000;
    color: #fff;
    border: 1px solid #d4af37;
    border-radius: 6px;
    padding: 10px;
    font-size: 1em;
    outline: none;
    transition: 0.3s;
    caret-color: #d4af37;
}

input:focus,
select:focus {
    border-color: #b9972e;
}

.login-btn {
    margin-top: 15px;
    background: #d4af37;
    color: #000;
    font-weight: bold;
```

```
    padding: 10px;
    border: none;
    border-radius: 6px;
    font-size: 1em;
    cursor: pointer;
    transition: .3s;
}

.login-btn:hover {
    background: #c2a12e;
}

.links {
    margin-top: 12px;
    text-align: center;
    font-size: 0.85em;
}

.links a {
    color: #b2962d;
    text-decoration: none;
    margin: 0 6px;
    transition: .3s;
}

.links a:hover {
    color: #fff;
}

.tips-section {
    margin-top: 18px;
```

```
    padding-top: 12px;
    border-top: 1px solid rgba(212, 175, 55, 0.3);
    color: #d4af37;
    text-align: center;
    font-size: 0.9rem;
    letter-spacing: 1px;
    text-transform: uppercase;
    cursor: pointer;
    user-select: none;
}

.tips-section span {
    display: block;
    font-size: 0.85rem;
    margin-top: 6px;
    color: #ffdb58;
    opacity: 0;
    transition: opacity 1.2s ease;
    font-weight: 600;
}

.tips-section:hover strong {
    color: #ffdb58;
    transition: color 0.3s ease;
}

footer {
    position: absolute;
    bottom: 12px;
    width: 100%;
    text-align: center;
```

```
            color: #777;
            font-size: 0.75em;
            letter-spacing: 0.5px;
        }
    </style>
</head>

<body>
    <canvas id="network"></canvas>

    <div class="container">
        <div class="logo">
            <img src="https://i.postimg.cc/xjy79m1t/Whats-App-Image-2025-10-12-at-13-32-07-
54be146c.jpg" alt="A.C.E Logo">
        </div>
        <div class="title">Create Account</div>
        <div class="byline">Join the A.C.E Journey</div>

        <div id="error-message"></div>

        <form id="registerForm">
            <label for="name">Full Name</label>
            <input id="name" type="text" placeholder="Enter your name" required>

            <label for="email">Email</label>
            <input id="email" type="email" placeholder="Enter your email" required>

            <label for="password">Password</label>
            <input id="password" type="password" placeholder="Create a password" required>

            <label for="role">Select Role</label>
```

```
      <select id="role" required>
    <option value="">Select...</option>
    <option value="Student">Student</option>
    <option value="Teacher">Teacher</option>
    <option value="Parent">Parent</option>
  </select>

      <button type="submit" class="login-btn">Sign Up</button>
    </form>

    <div class="links">
      Already have an account? <a href="index.html">Login</a>
    </div>

    <div class="tips-section" id="tips-section">
      <strong>TIPS:</strong>
      <span id="tip-text"></span>
    </div>
  </div>

  <footer>© 2025 A.C.E Initiative | Concept Prototype</footer>

  <script>
    // === Registration Submit ===
    const form = document.getElementById("registerForm");
    const msg = document.getElementById("error-message");

    form.addEventListener("submit", async(e) => {
      e.preventDefault();
      msg.textContent = "⏳ Registering...";
      msg.style.color = "#ffdb58";
```

```
const name = document.getElementById("name").value.trim();
const email = document.getElementById("email").value.trim();
const password = document.getElementById("password").value.trim();
const role = document.getElementById("role").value;

try {
  const res = await fetch("/api/auth/register", {
    method: "POST",
    headers: {
      "Content-Type": "application/json"
    },
    body: JSON.stringify({
      name,
      email,
      password,
      role
    }),
  });
  const data = await res.json();

  if (res.ok) {
    msg.style.color = "#00cc66";
    msg.textContent = "✅ Registration successful!";
    setTimeout(() => (window.location.href = "index.html"), 1500);
  } else {
    msg.style.color = "#ff4d4d";
    msg.textContent = data.message || "❌ Registration failed";
  }
} catch (err) {
  msg.style.color = "#ff4d4d";
```

```
        msg.textContent = "❌ Server error. Please try again.";
    }
});

// === Background Network (same as index.html) ===
const canvas = document.getElementById("network");
const ctx = canvas.getContext("2d");
let w, h, particles = [];

function resize() {
    w = canvas.width = window.innerWidth;
    h = canvas.height = window.innerHeight;
}
window.addEventListener("resize", resize);
resize();

class Particle {
    constructor() {
        this.x = Math.random() * w;
        this.y = Math.random() * h;
        this.vx = (Math.random() - 0.5) * 0.3;
        this.vy = (Math.random() - 0.5) * 0.3;
        this.size = 1.5;
    }
    move() {
        this.x += this.vx;
        this.y += this.vy;
        if (this.x < 0 || this.x > w) this.vx *= -1;
        if (this.y < 0 || this.y > h) this.vy *= -1;
    }
    draw() {
```

```
        ctx.beginPath();
        ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
        ctx.fillStyle = "rgba(255,215,120,0.9)";
        ctx.shadowBlur = 10;
        ctx.shadowColor = "rgba(255,215,120,0.7)";
        ctx.fill();
        ctx.shadowBlur = 0;
    }
}

for (let i = 0; i < 60; i++) particles.push(new Particle());

function connect() {
    for (let a = 0; a < particles.length; a++) {
        for (let b = a + 1; b < particles.length; b++) {
            const dx = particles[a].x - particles[b].x;
            const dy = particles[a].y - particles[b].y;
            const dist = Math.sqrt(dx * dx + dy * dy);
            if (dist < 140) {
                const opacity = 0.35 * (1 - dist / 140);
                ctx.strokeStyle = `rgba(255,215,100,${opacity})`;
                ctx.lineWidth = 0.9;
                ctx.shadowBlur = 6;
                ctx.shadowColor = "rgba(255,215,120,0.9)";
                ctx.beginPath();
                ctx.moveTo(particles[a].x, particles[a].y);
                ctx.lineTo(particles[b].x, particles[b].y);
                ctx.stroke();
                ctx.shadowBlur = 0;
            }
        }
```

```
    }
}

function animate() {
    ctx.clearRect(0, 0, w, h);
    particles.forEach(p => {
        p.move();
        p.draw();
    });
    connect();
    requestAnimationFrame(animate);
}
animate();

// === Tips ===
const tips = [
    "LEARN SOMETHING NEW EVERY DAY",
    "CURIOSITY CREATES INNOVATION",
    "PROGRESS BEATS PERFECTION",
    "THINK. CREATE. EVOLVE.",
    "STAY FOCUSED AND CONSISTENT",
    "EFFORT BUILDS EXCELLENCE"
];
const tipText = document.getElementById("tip-text");
let tipIndex = 0;

function showNextTip() {
    tipText.style.opacity = 0;
    setTimeout(() => {
        tipText.textContent = tips[tipIndex];
        tipText.style.opacity = 1;
```

```
            tipIndex = (tipIndex + 1) % tips.length;
        }, 500);
    }
    showNextTip();
    setInterval(showNextTip, 6000);
  </script>
</body>

</html>

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>A.C.E | Signup</title>
  <style>
    body {
        background: #000;
        color: #d4af37;
        font-family: "Georgia", serif;
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
    }

    .signup-container {
        background: rgba(0, 0, 0, 0.85);
        padding: 30px 40px;
```

```
    border-radius: 12px;
    border: 1px solid #d4af37;
    width: 350px;
    box-shadow: 0 0 20px rgba(212, 175, 55, 0.3);
}

h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #ffdb58;
}

label {
    font-weight: bold;
    display: block;
    margin: 10px 0 4px;
}

input,
select {
    width: 100%;
    padding: 10px;
    border-radius: 6px;
    border: 1px solid #d4af37;
    background: #000;
    color: #ffdb58;
    outline: none;
    font-size: 1em;
}

button {
```

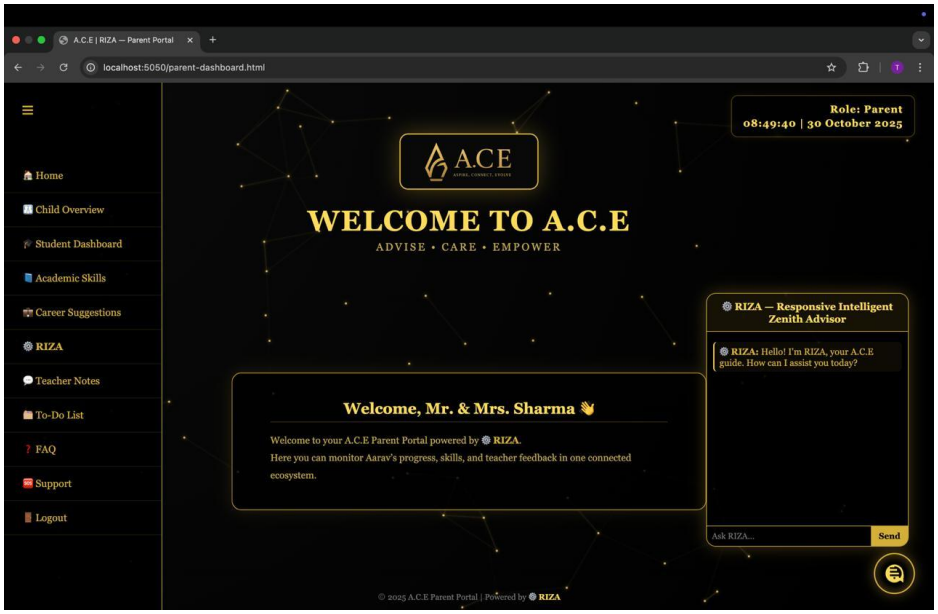
```
            width: 100%;
            margin-top: 15px;
            background: #d4af37;
            color: #000;
            border: none;
            padding: 10px;
            border-radius: 6px;
            font-weight: bold;
            cursor: pointer;
            transition: 0.3s;
        }

        button:hover {
            background: #ffdb58;
        }

        #message {
            margin-top: 15px;
            text-align: center;
            color: #ff4d4d;
            font-size: 0.9em;
            min-height: 20px;
        }
    </style>
</head>

<body>
    <div class="signup-container">
        <h2>Sign Up</h2>

        <label for="name">Name</label>
```

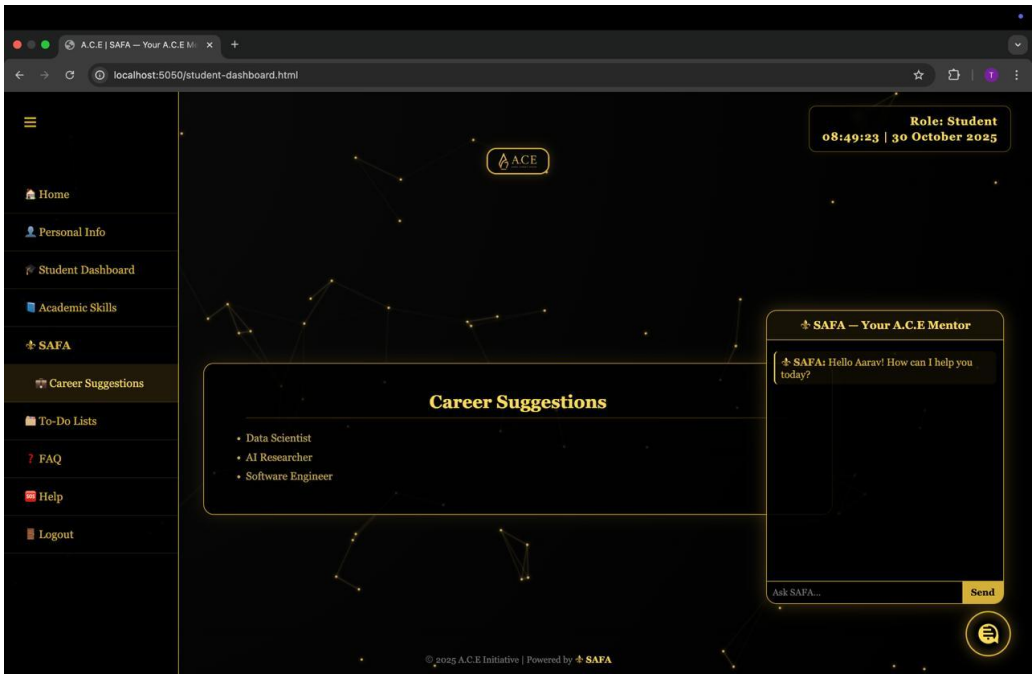
```
    <input type="text" id="name" placeholder="Enter your name" required>

    <label for="email">Email</label>
    <input type="email" id="email" placeholder="Enter your email" required>

    <label for="password">Password</label>
    <input type="password" id="password" placeholder="Enter your password" required>

    <label for="role">Role</label>
    <select id="role">
  <option value="Student">Student</option>
  <option value="Teacher">Teacher</option>
  <option value="Parent">Parent</option>
</select>

    <button type="button" onclick="signupUser()">Sign Up</button>
    <div id="message"></div>
</div>

<script>
    async function signupUser() {
        const name = document.getElementById("name").value.trim();
        const email = document.getElementById("email").value.trim();
        const password = document.getElementById("password").value.trim();
        const role = document.getElementById("role").value;
        const messageEl = document.getElementById("message");

        // Simple validation
        if (!name || !email || !password) {
            messageEl.textContent = "   All fields are required.";
            return;
```

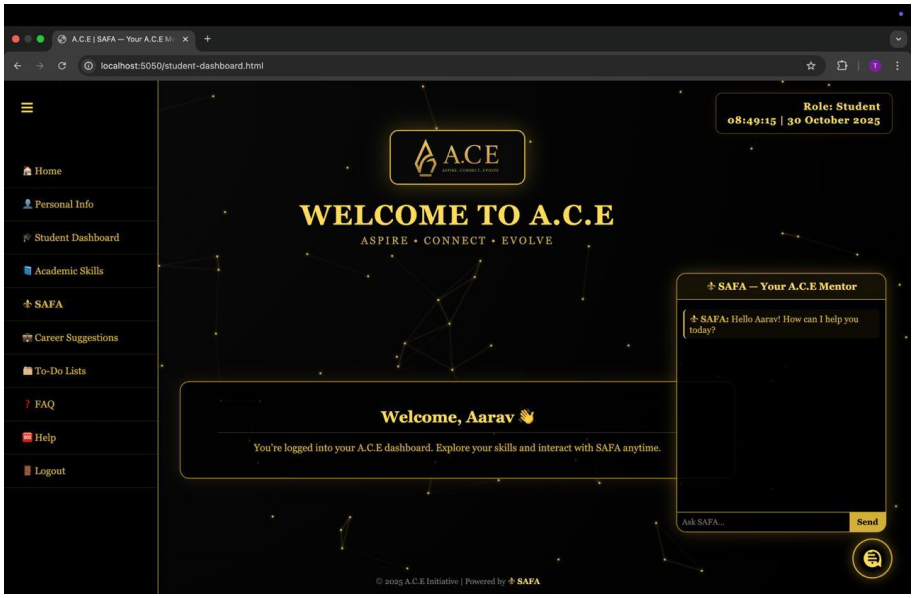
```
}

try {
    const response = await fetch("http://localhost:5050/api/users/signup", {
        method: "POST",
        headers: {
            "Content-Type": "application/json"
        },
        body: JSON.stringify({
            name,
            email,
            password,
            role
        })
    });

    const data = await response.json();

    if (response.ok) {
        messageEl.style.color = "#4CAF50";
        messageEl.textContent = "✅ Signup successful! Redirecting...";
        setTimeout(() => {
            window.location.href = "login.html"; // redirect after signup
        }, 1500);
    } else {
        messageEl.style.color = "#ff4d4d";
        messageEl.textContent = data.message || "❌ Signup failed.";
    }
} catch (error) {
    console.error(error);
    messageEl.textContent = "⚠ Could not connect to server.";
```

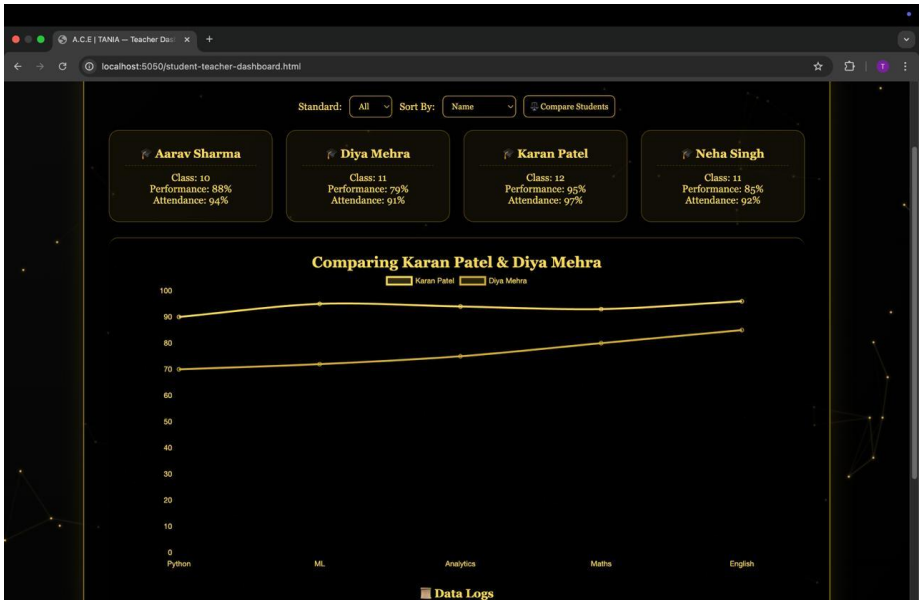
```
        }
      }
  </script>
</body>

</html>
```

Screenshots and Output



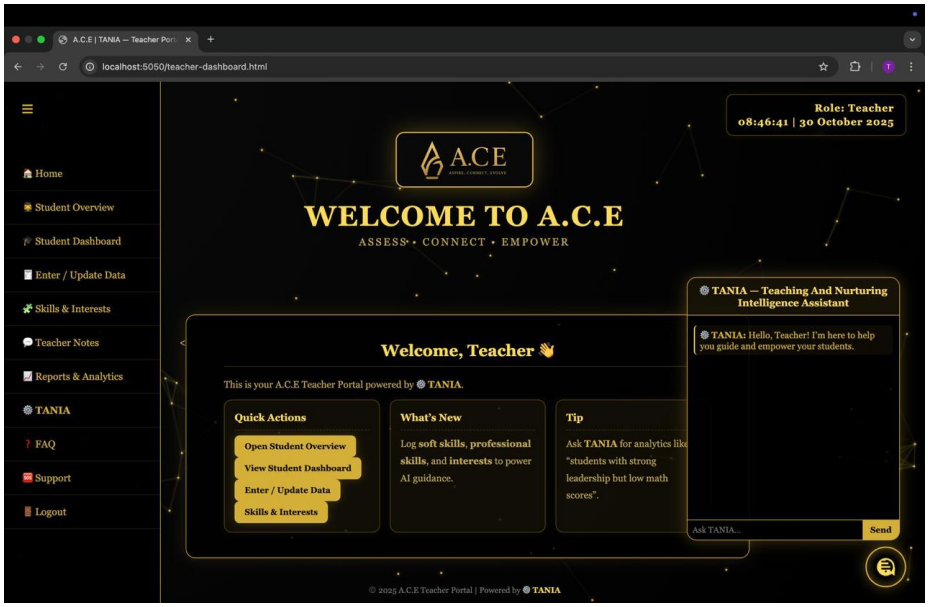


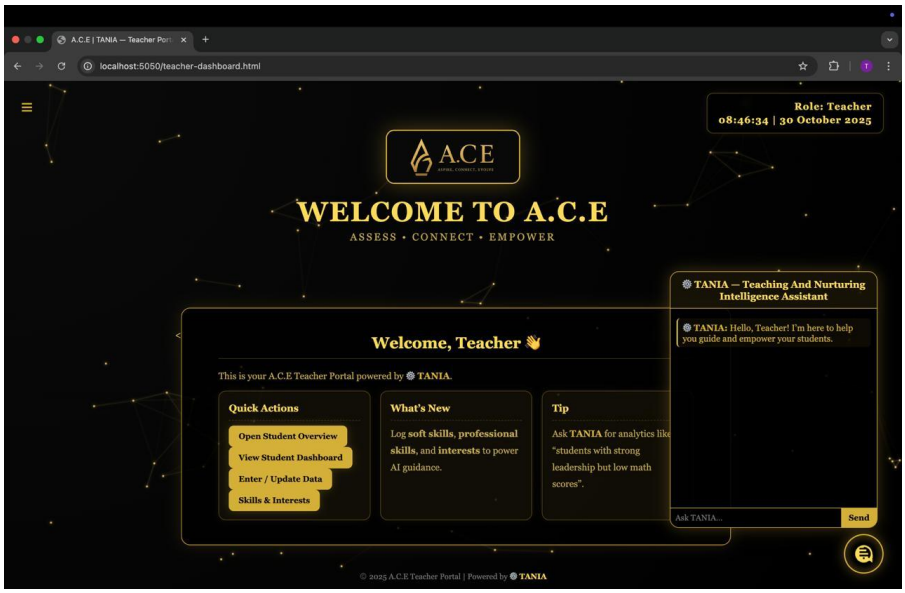

A.C.E | SAFA — Your A.C.E M
localhost:5050/student-dashboard.html
Role: Student
08:49:15 | 30 October 2025
A.C.E
WELCOME TO A.C.E
ASPIRE • CONNECT • EVOLVE
Home
Personal Info
Student Dashboard
Academic Skills
SAFA
Career Suggestions
To-Do Lists
FAQ
Help
Logout
SAFA — Your A.C.E Mentor
SAFA: Hello Aarav! How can I help you today?
Welcome, Aarav 👋
You're logged into your A.C.E dashboard. Explore your skills and interact with SAFA anytime.
Ask SAFA...
Send
© 2025 A.C.E Initiative | Powered by SAFA

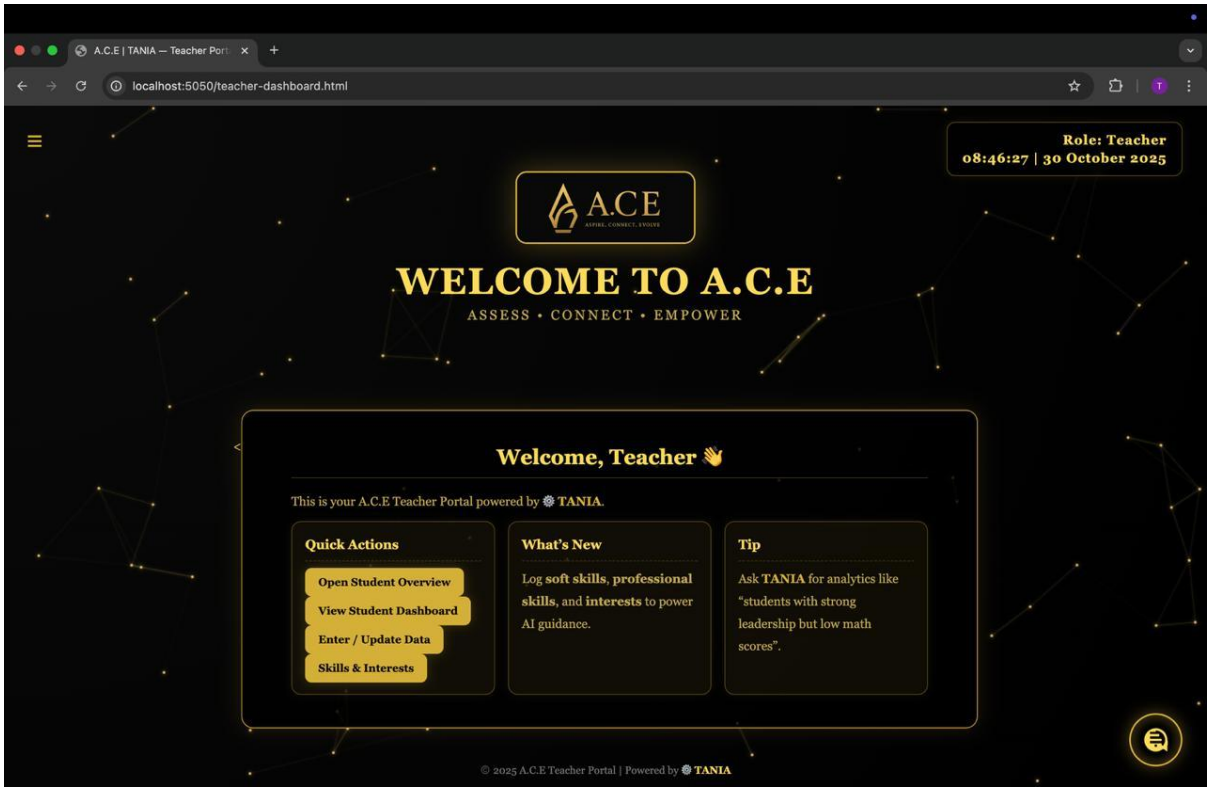

A.C.E | TANIA — Teacher Das
localhost:5050/student-teacher-dashboard.html
Standard: All
Sort By: Name
Compare Students
Aarav Sharma
Class: 10
Performance: 88%
Attendance: 94%
Diya Mehra
Class: 11
Performance: 79%
Attendance: 91%
Karan Patel
Class: 12
Performance: 95%
Attendance: 97%
Neha Singh
Class: 11
Performance: 85%
Attendance: 92%
Comparing Karan Patel & Diya Mehra
Karan Patel
Diya Mehra
100
90
80
70
60
50
40
30
20
10
0
Python
ML
Analytics
Maths
English
Data Logs

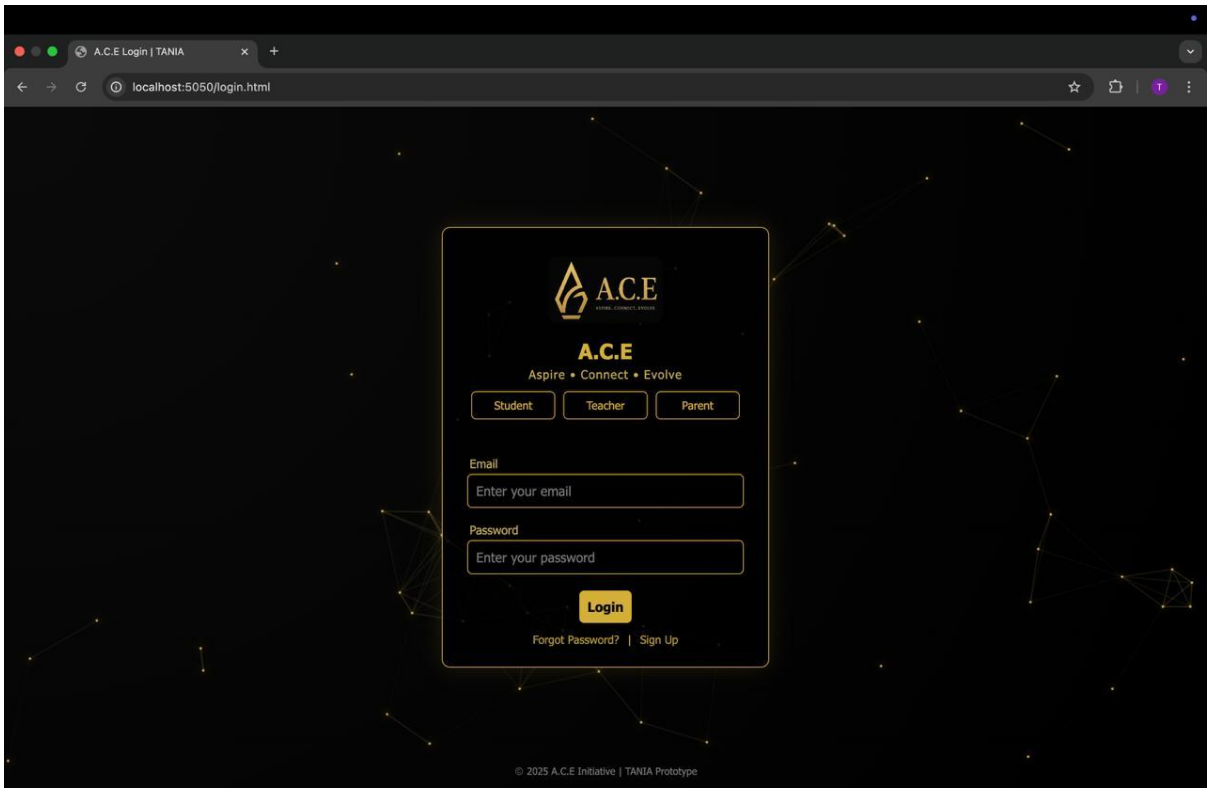
A.C.E | TANIA — Teacher Das
localhost:5050/student-teacher-dashboard.html
Aarav Sharma
Class: 10
Performance: 88%
Attendance: 94%
Diya Mehra
Class: 11
Performance: 79%
Attendance: 91%
Karan Patel
Class: 12
Performance: 95%
Attendance: 97%
Neha Singh
Class: 11
Performance: 85%
Attendance: 92%
Diya Mehra — Class 11
Current
100
90
80
70
60
50
40
30
20
10
0
Python
ML
Analytics
Maths
English
Old Stats
New Stats

A.C.E | TANIA — Teacher Das
localhost:5050/student-teacher-dashboard.html
Aarav Sharma
Class: 10
Performance: 88%
Attendance: 94%
Diya Mehra
Class: 11
Performance: 79%
Attendance: 91%
Karan Patel
Class: 12
Performance: 95%
Attendance: 97%
Neha Singh
Class: 11
Performance: 85%
Attendance: 92%
Diya Mehra — Class 11
Current
100
90
80
70
60
50
40
30
20
10
0
Python
ML
Analytics
Maths
English
Old Stats
New Stats


A.C.E | TANIA — Teacher Port
localhost:5050/teacher-dashboard.html
Role: Teacher
08:46:41 | 30 October 2025
Home
Student Overview
Student Dashboard
Enter / Update Data
Skills & Interests
Teacher Notes
Reports & Analytics
TANIA
FAQ
Support
Logout
A.C.E
WELCOME TO A.C.E
ASSESS • CONNECT • EMPOWER
TANIA — Teaching And Nurturing Intelligence Assistant
TANIA: Hello, Teacher! I'm here to help you guide and empower your students.
Welcome, Teacher 👋
This is your A.C.E Teacher Portal powered by TANIA.
Quick Actions
Open Student Overview
View Student Dashboard
Enter / Update Data
Skills & Interests
What's New
Log soft skills, professional skills, and interests to power AI guidance.
Tip
Ask TANIA for analytics like "students with strong leadership but low math scores".
Ask TANIA...
Send
© 2025 A.C.E Teacher Portal | Powered by TANIA


A.C.E | TANIA — Teacher Port
localhost:5050/teacher-dashboard.html
Role: Teacher
08:46:34 | 30 October 2025
A.C.E
WELCOME TO A.C.E
ASSESS • CONNECT • EMPOWER
TANIA — Teaching And Nurturing Intelligence Assistant
TANIA: Hello, Teacher! I'm here to help you guide and empower your students.
Welcome, Teacher 👋
This is your A.C.E Teacher Portal powered by TANIA.
Quick Actions
Open Student Overview
View Student Dashboard
Enter / Update Data
Skills & Interests
What's New
Log soft skills, professional skills, and interests to power AI guidance.
Tip
Ask TANIA for analytics like "students with strong leadership but low math scores".
Ask TANIA...
Send
© 2025 A.C.E Teacher Portal | Powered by TANIA


A.C.E | TANIA — Teacher Port
localhost:5050/teacher-dashboard.html
Role: Teacher
08:46:27 | 30 October 2025
A.C.E
WELCOME TO A.C.E
ASSESS • CONNECT • EMPOWER
Welcome, Teacher 👋
This is your A.C.E Teacher Portal powered by TANIA.
Quick Actions
Open Student Overview
View Student Dashboard
Enter / Update Data
Skills & Interests
What's New
Log soft skills, professional skills, and interests to power AI guidance.
Tip
Ask TANIA for analytics like "students with strong leadership but low math scores".
© 2025 A.C.E Teacher Portal | Powered by TANIA


A.C.E Login | TANIA
localhost:5050/login.html
A.C.E
A.C.E
Aspire • Connect • Evolve
Student
Teacher
Parent
Email
Enter your email
Password
Enter your password
Login
Forgot Password? | Sign Up
© 2025 A.C.E Initiative | TANIA Prototype

10. Testing

Testing is an essential phase in the software development lifecycle (SDLC) used to ensure that the application performs according to its functional and non-functional requirements. The **A.C.E. (Aspire, Connect, Evolve)** web application underwent multiple stages of testing — including **unit testing, integration testing, functional testing, and user acceptance testing (UAT)** — to verify its accuracy, stability, usability, and performance.

10.1 Test Cases

Test Case ID	Test Scenario / Module	Test Description	Input / Action	Expected Result	Actual Result	Status
TC-01	**User Registration**	Verify if a new user can successfully create an account.	Enter valid name, email, password, and confirm password.	Account should be created and confirmation message displayed.	Account successfully created and verified via email.	✅ Working Correctly
TC-02	**Login Authentication**	Test user login with correct and incorrect credentials.	Enter valid and invalid login details.	System should allow login for valid users and deny invalid access.	Login successful for valid credentials; error shown for invalid input.	✅ Working Correctly
TC-03	**Form Validation**	Check mandatory field validation in registration and feedback forms.	Leave one or more fields blank and submit form.	System should show "Field Required" error message.	Validation messages displayed correctly for all missing fields.	✅ Working Correctly
TC-04	**AI Career Recommendation**	Validate that the AI engine provides relevant recommendations.	Input user profile details (interests, skills, academics).	Career suggestions should align with user preferences.	Recommendations displayed accurately based on input data.	✅ Working Correctly
TC-05	**Search Colleges and Scholarships**	Verify database search and filtering functionality.	Search for a course or scholarship by keyword.	System should show accurate, verified results.	Results correctly displayed from verified database.	✅ Working Correctly

Test Case ID	Test Scenario / Module	Test Description	Input / Action	Expected Result	Actual Result	Status
TC-06	**Parental Dashboard Access**	Ensure parents can view their child's progress.	Parent logs in using linked student ID.	Dashboard should show student's reports and suggestions.	Parental dashboard loaded with real-time data.	✅ Working Correctly
TC-07	**Teacher Analytics Dashboard**	Verify teacher access to analytics data.	Teacher logs in to view class trends.	Graphs and statistics should display student progress.	Data visualization loaded successfully.	✅ Working Correctly
TC-08	**Fraud Detection Module**	Check system's ability to identify unverified institutions.	Input a fake or unverified institution link.	Warning or alert message should be generated.	Warning successfully displayed; record flagged as unverified.	✅ Working Correctly
TC-09	**Responsiveness and UI Testing**	Test how well the platform adapts to different devices.	Open site on mobile, tablet, and desktop.	Layout and navigation should adjust automatically.	Responsive design works properly on all devices.	✅ Working Correctly
TC-10	**Forgot Password Functionality**	Verify password reset via email.	Enter registered email and request password reset.	Reset link should be sent to registered email.	Password reset link successfully sent.	✅ Working Correctly
TC-11	**Server Connectivity Test**	Ensure connection between frontend, backend, and database.	Access dashboard data after login.	Data should load without error or timeout.	Connection stable; data loads quickly.	✅ Working Correctly
TC-12	**Data Security Test**	Validate that passwords and user data are encrypted.	Attempt unauthorized data access.	System should deny access and secure data.	Unauthorized access blocked successfully.	✅ Working Correctly
TC-13	**Feedback**	Test feedback	Fill out	Data should be	Feedback saved	✅

Test Case ID	Test Scenario / Module	Test Description	Input / Action	Expected Result	Actual Result	Status
	Submission	form submission and storage.	feedback form and submit.	stored and confirmation shown.	successfully and confirmation displayed.	Working Correctly
TC-14	**Logout Functionality**	Verify if user can safely log out.	Click "Logout" from dashboard.	Session should end and redirect to homepage.	Logout works correctly; session cleared.	✅ Working Correctly
TC-15	**Error Handling Test**	Check how system handles unexpected input or errors.	Enter invalid data or simulate server issue.	System should show appropriate error message without crashing.	Error messages handled gracefully.	✅ Working Correctly

---

10.2 Result of Testing

All critical modules of the **A.C.E. system** were tested successfully and performed as expected. During the early testing phases, minor issues such as **UI alignment errors, slow page load during AI processing, and incomplete validation messages** were identified. These issues were fixed during iterative debugging and optimization.

After successful integration and user acceptance testing, the final version of the application met all defined functional requirements. The platform is now **stable, secure, and user-friendly**, with consistent performance across devices and browsers.

10.3 Summary of Testing

Testing Type	Objective	Outcome
Unit Testing	Verify individual modules and functions.	All modules passed without major errors.
Integration Testing	Ensure seamless interaction between front-end, back-end, and database.	Data communication stable and accurate.
Functional Testing	Validate business logic and core features.	All features meet user and system requirements.
UI/UX Testing	Assess responsiveness, navigation, and	Interface intuitive and mobile-

Testing Type	Objective	Outcome
	accessibility.	compatible.
Security Testing	Check user authentication, data encryption, and access control.	All sensitive data protected; no vulnerabilities found.
User Acceptance Testing (UAT)	Confirm usability and satisfaction from test users.	Positive feedback; platform ready for deployment.

Result

The **A.C.E. (Aspire, Connect, Evolve)** project has successfully achieved its primary objective of developing a comprehensive, AI-powered web application that provides **personalized and data-driven career guidance** for students. The system effectively bridges the gap between traditional career counseling and modern digital solutions by integrating artificial intelligence, real-time data analytics, and verified educational resources into a single, accessible platform.

Through the implementation of **intelligent algorithms**, the system is capable of analyzing a student's academic performance, interests, and personality traits to recommend the most suitable career paths. The project also succeeded in building a **secure and user-friendly interface** that allows students, teachers, and parents to collaborate effectively in the career planning process.

The major achievements of this project include:

AI-Based Career Recommendation System:
The project successfully implemented an artificial intelligence model capable of suggesting accurate and relevant career options based on user inputs and global employment trends.

Interactive and Responsive Web Interface:
A.C.E. was developed using **HTML, CSS, JavaScript, React.js, and Bootstrap**, ensuring that the platform is responsive, accessible, and user-friendly across all devices.

Dynamic and Verified Database:
The system integrates a database of accredited institutions, verified courses, and scholarship programs, ensuring that students only access authentic and trusted sources of information.

Collaboration Dashboard for Stakeholders:
The platform enables teachers and parents to monitor student progress through dashboards that display performance analytics, recommendations, and reports.

Fraud Detection and Data Security:
The project implemented a basic **AI-based fraud detection feature** that flags unverified institutions and protects users from educational scams, enhancing the system's reliability.

Cloud-Ready and Scalable Design:
The web application was developed with a **modular architecture**, allowing for easy scalability and deployment on cloud platforms such as AWS or Azure.

User Testing and Validation:
Comprehensive testing was conducted for functionality, UI responsiveness, and security. The results demonstrated high system stability, data accuracy, and positive user feedback.

Real-Time Market Alignment:
The project successfully incorporated global employment data and skill demand trends into the recommendation system, ensuring that students are guided toward careers that align with future market opportunities.

Conclusion

- What you learned
- Improvement needed in future

What We Learned

The development of the **A.C.E. (Aspire, Connect, Evolve)** web application has been a highly enriching academic and technical learning experience. Through this project, we gained a deep understanding of how technology can be applied to solve real-world educational challenges, especially those related to **career awareness, employability, and digital equity**.

From the conceptualization stage to implementation, the project enabled us to explore the **complete software development lifecycle (SDLC)** — including requirement analysis, system design, database management, front-end and back-end development, and testing. We also learned the importance of teamwork, collaboration, and version control while working with modern tools such as **Visual Studio Code, GitHub, and Postman**.

A major learning outcome of this project was understanding how **Artificial Intelligence (AI)** and **Data Analytics** can be integrated into web applications to personalize user experiences. The process of developing the AI-based career recommendation engine taught us how data-driven models can guide meaningful decision-making for students. We also learned about the ethical aspects of AI — including data privacy, transparency, and fairness — which are essential in educational technology platforms.

Additionally, this project enhanced our skills in **front-end technologies** like HTML, CSS, Bootstrap, and React.js, as well as **back-end technologies** like Node.js, Express, and MongoDB/MySQL. We learned to manage cloud-based architecture and deploy scalable systems on modern hosting platforms.

Beyond technical knowledge, we gained valuable insight into the **social and educational impact** of technology. We understood that a digital platform like A.C.E. has the power to bridge gaps in career guidance, promote equality, and help students from diverse backgrounds make informed career choices. The project also taught us the importance of **user experience (UX)**, **accessibility**, and **data accuracy** in developing applications that serve large-scale educational communities.

Overall, working on A.C.E. strengthened our ability to combine technical innovation with social responsibility — a vital skill in building meaningful, human-centered technology solutions.

12.2 Improvements Needed in the Future

While the current version of **A.C.E.** successfully fulfills its objectives as an AI-powered career guidance platform, there is still considerable potential for enhancement and innovation in future iterations. Continuous improvement will ensure that the system remains **relevant, scalable, and adaptive** to evolving educational and market needs.

Enhanced AI Algorithms:
Future updates could include more advanced machine learning models such as deep neural networks for better prediction accuracy. These models can learn from larger datasets and provide even more precise career matches based on subtle behavioral patterns.

Mobile Application Development:
Although the current version is web-based and responsive, developing a dedicated **mobile application (Android and iOS)** would improve accessibility for students who primarily rely on smartphones for learning.

Integration with E-Learning Platforms:
The next version can integrate with **MOOCs (Massive Open Online Courses)** such as Coursera, edX, or Skillshare to automatically recommend certified skill-building programs based on each student's chosen career path.

Inclusion of AI Chatbot for Real-Time Support:
Adding an **AI-powered virtual assistant** would allow users to ask questions and receive instant guidance, thereby improving interactivity and user engagement.

Localization and Multilingual Support:
A.C.E. can be expanded to support **regional languages** to reach students in rural and non-English-speaking regions, ensuring inclusivity and equal access to all.

Expanded Analytics for Policymakers:
Future versions can include advanced **data visualization dashboards** for government bodies and educational institutions to monitor employment trends, regional skill shortages, and policy outcomes.

Gamification and Motivation Tools:
Introducing **badges, progress tracking, and reward systems** can motivate students to complete skill-building tasks and engage regularly with the platform.

Integration with Employer Networks:
Partnering with industry and corporate networks will allow A.C.E. to provide internship opportunities, job listings, and mentorship programs aligned with the recommended career paths.

Improved Data Security and Compliance:
As user data grows, strengthening **data protection mechanisms** and ensuring compliance with international privacy regulations such as GDPR will be essential for long-term trust and security.

AI Ethics and Bias Reduction:
Regular audits and testing should be performed to identify and eliminate potential bias in AI recommendations, ensuring fair and unbiased guidance for all users.

Reference

- Books, websites (in proper format)

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